Quasi-Zenith Satellite System
Interface Specification
DCX Service

(IS-QZSS-DCX-004)

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Cabinet Office

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Revision History

Rev. No.	Date	Page	Revision
001	March 29,2024	-	-
002	October 3,2025	11, 12, 26, 28, 29, 34, 52, 56, 58, 75	Correction of errors
		 7, 8, 49 21, 22, 23, 24, 25 59 69 69 70	Refinement of descriptions ・Sections 4.2.1. and 5.2. -Refined description about Service Area. ・Section 4.2.3.11. (1) -Refined description on A11-Guidance to react library field. -Moved Table 5.10-1 in section 5.10. to Table 4.2-15 in section 4.2.3.11. ・Section 5.4.2. -Refined description regarding J-Alert cancellation. ・Section 5.8. -Added description of how to specify the target area. ・Section 5.8.1. -Changed the section title. ・Section 5.8.2. -Changed the section title.
		 37,38,39,40 74	Allowed combined use of A11 and the Extended Message (EX2-EX7). ・Sections 4.2.4.1.2. – 4.2.4.1.7. -Removed unnecessary description. ・Section 5.10. -Revised the description of how to use the evacuation action instructions indicated in A11 and Extended -Removed Figure 5.2-1.
		22	Adding new Evacuation Action Instruction List. ・Section 4.2.3.11. -Added new Evacuation Action Instruction List of "This is a test message for DCX." in Table 4.2-15, which is intended to be used during the test.
003	March 28,2025	7,8,13,14,15,19,20, 21,22,35,36,37,38, 39,40,50,52,54,57, 68,74	Delete the [tentative name] notation from “municipality transmitted information” and named “ information from local government”. ・Section 4.2.1. ・Sections 4.2.3.1.-4.2.3.3. ・Sections 4.2.3.9.-4.2.3.11. ・Table 4.2-15 ・Section 4.2.3.17. ・Section 4.2.4.1. ・Sections 4.2.4.1.2.-4.2.4.1.3. ・Sections 4.2.4.1.4.-4.2.4.1.8. ・Section 5.2. ・Section 5.3.1. ・Section 5.4. ・Section 5.4.1. ・Section 5.7. ・Section 5.10.
		15,54,68	Change “Municipality” to “Local Government” of Provider Identifier definition for Japan ・Sections 4.3.3.3. ・Table 4.2-6 ・Sections 5.3.1 ・Sections 5.7.

Rev. No.	Date	Page	Revision
		37,41,42,45,69,70	Change “Municipality” to “city(s), town(s) and village(s)”. ・Section 4.2.4.1.1. ・Table 4.2-21 ・Section 4.2.4.2.1 ・Table 4.2 24 ・Section 4.2.4.2.2 ・Figure 4.2-8 ・Sections 5.8.1.-5.8.2.
004	May 29,2026		Following revisions made in conjunction with the update from Version 1.1 to Version 1.2 of the CAMF, the applicable document
		5, 6	Add Sections 2.4 “Definitions” and 2.5 “General Assumptions”
		Across the document	Change “empty” to “not used” or “reserved.” The definitions of “not used” and ‘reserved’ are provided in Section 2.4, “Definitions.”
		Across the document	Change “guidance” to “instruction”, to follow the CAP terminology. "Guidance" is softer than "instruction" and may be perceived as less urgent.
		Across the document	Change “display” to “notification”, to make it more generic since the alert might be notified not only with a display.
		Across the document	The consistent rounding rules below have been applied to all calculated values to ensure that the reported precision is uniform and appropriate for an accuracy of 1 meter: -Distances are rounded to the nearest 1 meter. -Latitude/longitude values are rounded to 1e-6 degrees. -Ellipse rotation angles are rounded to 1e-5 degrees.
		Across the document	Change “Selection of Library” and “guidance library” to “Type of Library”, in line with the CAMF update.
		14, 37, 49, 54, 56	Change “Main subject for Specific Settings” to “Type of Specific Settings”, in line with the CAMF update.
		20	Change “Day/Hour/Minute” to “WEEKDAY – Time [UTC]”. To clarify that “Day” refers to the day of the week (not the calendar date).
		20	Change “not used” or “reserved.” The definitions of “not used” and ‘reserved’ are provided in Section 2.4, “Definitions.”
		36, 41	Add a definition of “Azimuth angle” to avoid confusion since the CAMF convention is not the standard convention in navigation.
		74	Change “Geographic coordinates” to “geodetic coordinates.” This is because ‘geodetic’ is a precise term referring to the Earth ellipsoid (WGS84), whereas “geographic” is ambiguous and could refer to the Earth as a sphere.
		79	Add “Geo-Fencing,” which has been added to the CAMF, as Section 5.12

Table of Contents

1. Scope

This document describes the interface specifications of the satellite report for disaster and crisis management (Extended Information) (DCX) between the space segment of QZSS and the user segment. The interface specifications described herein include message specifications.

2. Relevant Documents and Definition of Terms

2.1. Applicable Documents

The following documents constitute part of this document within the scope defined in this document.

(1) Quasi-Zenith Satellite System Interface Specification Sub-meter Level Augmentation Service (IS-QZSS-L1S)

(2) Emergency Warning Satellite Service COMMON ALERT MESSAGE FORMAT SPECIFICATION

(3) Identification code for cities， towns and villages (JIS X 0402)

2.2. Reference Documents

There are no reference documents.

2.3. Abbreviations

-A-

-B-

-C-

CRC	Cyclic Redundancy Check

-D-

DCX	Satellite Report for Disaster and Crisis Management (Extended Information)

-E-

EWSS	Emergency Warning Satellite Service

-F-

-G-

-H-

-I-

-J-

-K-

-L-

-M-

MT	Message Type

-N-

-O-

-P-

PAB	Preamble

-Q-

QZSS	Quasi-Zenith Satellite System

-R

-S-

SLAS	Sub-meter Level Augmentation Service

-T-

-U-

UTC	Coordinated Universal Time

-V-

-W-

-X-

-Y-

-Z-

2.4. Definition

For the purposes of the format specification described in this document, the following terms shall have the meanings defined below:

Emergency Warning Message (EWM): The 122-bit code containing the warning information that is transmitted by satellite.

Receiver: A device capable of receiving and processing EWM.

Instruction: A recommended action provided to the public to reduce risk or harm during an emergency. Instruction may also appear under the terms “guidance” or “guidance to react” in other EWSS documentation.

Library: A collection of codes, where each code is assigned a specific value and meaning, used to ensure complete and consistent encoding, transmission, and interpretation of the EWM. The library includes lists of instructions, but it is not limited to them.

Azimuth angle: The angle between a fixed reference direction (e.g. a cardinal point) and a target, measured in the horizontal plane.

Not used: A field value that is intentionally assigned to indicate the absence of value. Not used values may be encoded by senders and shall be treated by receivers as a field with no value.

Reserved: A field value that is intentionally not assigned. Reserved values shall not be encoded by senders and shall be ignored or treated as invalid by receivers. Reserved values are allocated to support future evolution of the format.

2.5. General Assumptions

Unless otherwise specified, the following general assumptions apply to the format specification described in this document:

Reference frame: All positions, coordinates, and spatial references used in this document are expressed in the WGS84 geodetic reference frame.

Time System: All times in this document are expressed in UTC.

Units: All angular measurements are expressed in degrees, and all distances are expressed in kilometres. Units are recalled in formulas and tables between square brackets, e.g. *[degrees]*, *[kilometres]* …

Numerical format: All numerical values use the dot "." as the decimal separator and are rounded as follows: 10^{-6} degrees for latitudes and longitudes, 10^{-5} degrees for angles, and 1 meter for distances.

3. Signal Specifications

DCX messages are transmitted by L1S signals as part of sub-meter level augmentation service (SLAS).

See Chapter 3 of the applicable document (1) for information about the signal specifications of L1S signals.

4. Message Specifications

4.1. Message Configuration

4.1.1. Overview

DCX messages are transmitted by L1S signals as part of the SLAS message.

See chapter 4.1.1.1 of the applicable document (1) for information about the message block format of the L1S signal.

4.1.2. Timing

DCX message are transmitted by L1S signals as part of the SLAS message.

See chapter 4.1.1.2 of the applicable document (1) for information about the transmission interval.

4.1.3. Cyclic Redundancy Check (CRC)

DCX messages are transmitted by L1S signals as part of the SLAS message.

See chapter 4.1.1.3 of the applicable document (1) for information about the cyclic redundancy check (CRC).

4.1.4. Transmission Order of DCX Messages

(1) Message transmission for Japan

DCX messages are transmitted based on the priority of the message. The priority is determined according to a combination of A1-Message Type and A5-Severity in Message Type 44 (See 4.2.3.1for information about parameter A1-Message Type. See 4.2.3.5for information about parameter A5-Severity.)

Table 4.1-1 shows the message priority.

Table 4.1-1 Message Priority

A1-Message Type / **A5-Severity**	00 Test	01 Alert	10 Update	11 All Clear
00 Unknown	Priority 6	Priority 3	Priority 3	Priority 3
01 Moderate-Possible threat to life or property	Priority 6	Priority 3	Priority 3	Priority 3
10 Severe-Significant threat to life or property	Priority 5	Priority 2	Priority 2	Priority 3
11 Extreme-Extraordinary threat to life or property	Priority 4	Priority 1	Priority 1	Priority 3

Priority High ←→ Low

Priority 1 Priority 2 Priority 3 Priority 4 Priority 5 Priority 6

Each DCX message is typically transmitted twice. However, the number of repetition of these transmissions can be modified through parameter adjustments, indicating the possibility of future changes.

Messages with higher priority may interrupt this sequence and be transmitted first.

Due to such interruptions, the same message may not always be transmitted consecutively.

(2) Message transmission for use outside Japan

The transmission order of DCX messages is determined by the order in which they are received from outside Japan, with each message being transmitted as many times as received.

4.2. Message Contents

4.2.1. Overview

Message type 44 (MT44) is used to transmit DCX message (L-Alert), DCX message (J-Alert), DCX message (information from local government), and DCX message (information from organizations outside Japan).

DCX message (L-Alert) is created based on the evacuation information transmitted by the Foundation for MultiMedia Communications.

DCX message (J-Alert) is created based on the national protection information transmitted by the Fire and Disaster Management Agency and other national protection information.

DCX message (information from local government) is created based on the evacuation information transmitted by the local government.

DCX message (information from organizations outside Japan) is created based on the disaster organization control information received from organizations outside Japan.

Table 4.2-1 shows the information transmitted as MT44.

DCX message (L-Alert), DCX message (J-Alert) and DCX message (information from local government) are transmitted for users in Japan.

DCX message (Information from organizations outside Japan) is transmitted for users outside Japan.

See 5.2. for service areas of each service type.

Table 4.2-1 Information Transmitted as MT44

Service type	Transmitted Information	Information Source Controlled by	Contents Transmitted
For Japan	DCX message (L-Alert)	Foundation for MultiMedia Communications	Information created based on L-Alert (disaster information) received from the Foundation for MultiMedia Communications
	DCX message (J-Alert)	Fire and Disaster Management Agency, Ministry of Internal Affairs and Communications	Information created based on J-Alert (national protection information) received from the Fire and Disaster Management Agency, Ministry of Internal Affairs and Communications
	DCX message (information from local government)	Local Government	Information created based on evacuation information received from the local government
For use outside Japan	DCX message (Information from organizations outside Japan)	Disaster prevention organization outside Japan	Information created based on disaster information received from organizations outside Japan

In the format of MT44, the 10-bit Satellite Designation (SD), 122-bit Common Alert Message Format (CAMF), and 74-bit Extended Message fields follow the MT field.

Figure 4.2-1 shows the format of MT44. See the applicable document (2) for details on CAMF.

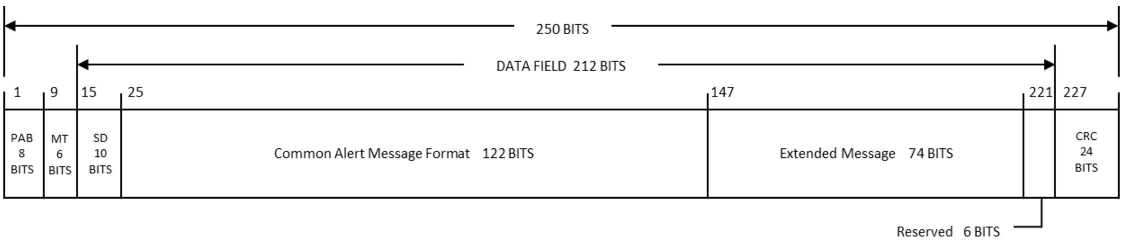


Figure 4.2-1 MT44 Format

4.2.2. Satellite Designation (SD)

This field consists of the service type indicating which QZSS satellite transmits MT44 for Japan and which QZSS satellite transmits MT44 for use outside Japan and the information notifying the receiver about which QZSS satellite is transmitting MT44. Table 4.2-2 shows the field format. See 5.6 for details.

Table 4.2-2 Satellite Designation (SD) Field Format

Parameter	Name	Number of Bits	Description
SDMT	Satellite Designation Mask Type	1	Satellite designation information mask type Indicates the type of SDM (satellite designation information mask), which is one of the following. "0": Service type. SDM indicates the service type indicating whether MT44 is for Japan or for use outside Japan. "1": MT44 transmission status. SDM indicates the MT44 transmission status. * The "MT44 transmission status" is transmitted infrequently, such as once a minute.
SDM	Satellite Designation Mask	9	Satellite designation information mask Each bit indicates the satellite designation information of one of the 9 satellites. When SDMT is "0", the bit indicates the service type of the corresponding satellite. "0": For Japan "1": For use outside Japan When SDMT is "1", the bit indicates the MT44 transmission status of the corresponding satellite. "0": Transmission stopped "1": Transmission in progress
Bit_1	Satellite (1)	1	Satellite designation information of Satellite (1)
Bit_2	Satellite (2)	1	Satellite designation information of Satellite (2)
Bit_3	Satellite (3)	1	Satellite designation information of Satellite (3)
Bit_4	Satellite (4)	1	Satellite designation information of Satellite (4)
Bit_5	Satellite (5)	1	Satellite designation information of Satellite (5)
Bit_6	Satellite (6)	1	Satellite designation information of Satellite (6)

Parameter		Name	Number of Bits	Description
	Bit_7	Satellite (7)	1	Satellite designation information of Satellite (7)
	Bit_8	Satellite (8)	1	Satellite designation information of Satellite (8)
	Bit_9	Satellite (9)	1	Satellite designation information of Satellite (9)

4.2.3. Common Alert Message Format (CAMF)

Figure 4.2-2 shows the message format of the Common Alert Message Format. Table 4.2-3 shows the parameter definition. Details for each field are presented starting from section 4.2.3.1. The tables within each section outline the contents of these fields and are designed to aid user understanding. It should be noted that some tables only illustrate a part of the contents as examples. See the applicable document (2) for details on CAMF.

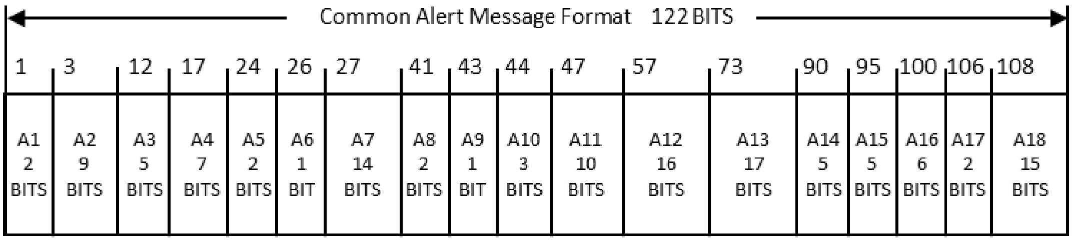


Figure 4.2-2 Common Alert Message Format (CAMF) Message Format

Table 4.2-3 Common Alert Message Format (CAMF) Parameter Definition

Parameter	Description	Number of Bits
A1	Message Type → See 4.2.3.1	2
A2	Country/Region Name → See 4.2.3.2	9
A3	Provider Identifier → See 4.2.3.3	5
A4	Hazard Category and Type → See 4.2.3.4	7
A5	Severity → See 4.2.3.5	2
A6	Hazard Onset: Week Number → See 4.2.3.6	1
A7	Hazard Onset: Time of the Week → See 4.2.3.7	14
A8	Hazard Duration → See 4.2.3.8	2
A9	Type of Library → See 4.2.3.9	1
A10	Version of Library → See 4.2.3.10	3
A11	Instruction library → See 4.2.3.11	10
A12	Ellipse Centre Latitude → See 4.2.3.12	16
A13	Ellipse Centre Longitude → See 4.2.3.13	17
A14	Ellipse Semi-Major Axis → See 4.2.3.14	5

Parameter	Description	Number of Bits
A15	Ellipse Semi-Minor Axis → See 4.2.3.15	5
A16	Ellipse Azimuth → See 4.2.3.16	6
A17	Type of Specific Settings → See 4.2.3.17	2
A18	Specific Settings → See 4.2.3.18	15

4.2.3.1. A1-Message Type

This field is a 2-bit code indicating the message type.

The message types are Test, Alert, Update, and All Clear.

Table 4.2-4 shows the message types.

Table 4.2-4 Message Type

Code [2bits]	Message Type
00	Test
01	Alert
10	Update
11	All Clear

“00” (Test) indicates test transmission. It is used for training and system testing on the user receiver side.

“01” (Alert) indicates the occurrence of an alert.

“10” (Update) indicates that a field other than A2-Country/Region Name, A3-Provider Identifier, and A4-Hazard Category and Type has been updated. For DCX message (L-Alert), DCX message (J-Alert), and DCX message (information from local government), an alert is identified according to the following conditions.

・DCX message (L-Alert), DCX message (information from local government)

The values of the A2, A3, A4, and EX1 fields match.

・DCX message (J-Alert)

The values of the A2, A3, and A4 fields match.

“11” (All Clear) indicates the end of an alert. For DCX message (L-Alert), DCX message (J-Alert), and DCX message (information from local government), an alert is identified in the same way as “10” (Update).

4.2.3.2. A2-Country/Region Name

This field is a 9-bit code that indicates the country or region issuing the alert.

Table 4.2-5 shows the contents of Country/Region Name.

In the case of DCX message (L-Alert), DCX message (J-Alert), and DCX message (information from local government), the code is fixed to "001101111" indicating Japan.

Table 4.2-5 Country/Region Name

Code [9 bits]	Country Name
000000000	Afghanistan
000000001	Albania
000000010	Antarctica
・・・	・・・
001101111	Japan
・・・	・・・
011111000	Zambia
011111001	*reserved*
・・・	・・・
111111111	*reserved*

4.2.3.3. A3-Provider Identifier

This field is a 5-bit code that indicates the organization issuing the alert message.

Table 4.2-6 shows the contents of Provider Identifier.

In the case of DCX message (L-Alert), the code is fixed to "00001" (FMMC).

In the case of DCX message (J-Alert), "00010" (FDMA) or "00011" (Related Ministries) is specified.

In the case of DCX message (information from local government), the code is fixed to "00100" (Local Government).

Table 4.2-6 Provider Identifier

Code [5 bits]	Provider ID	Provider Name (Japan)	Provider Name (Australia)	Provider Name (Fiji)	Provider Name (Thailand)
00000	0	not used	not used	not used	not used
00001	1	Foundation for MultiMedia Communications (FMMC)	National Emergency Management Agency (NEMA)	National Disaster Management Office (NDMO)	Department of Disaster Prevention and Mitigation (DDPM)
00010	2	Fire and Disaster Management Agency (FDMA)	Bureau of Meteorology (BOM)	Fiji Meteorological Service (FMS)	Thai Meteorological Department (TMD)
00011	3	Related Ministries	Australian Climate Service (ACS)	Hydrology Section, Fiji Water Authority (FWA)	National Disaster Warning Center (NDWC)
00100	4	Local Government	Geoscience Australia (GA)	Mineral Resources Department (MRD)	Department of Mineral Resources (DMR)
00101	5	*reserved*	Commonwealth Scientific and Industrial Research Organisation (CSIRO)	Fiji Broadcasting Corporation (FBC)	Navy Hydrographic Department, Royal Thai Navy (RTN)
00110	6	*reserved*	Australian Bureau of Statistics (ABS)	*reserved*	Department of Water Resources (DWR)
00111	7	*reserved*	Resilience New South Wales (Resilience NSW)	*reserved*	Royal Irrigation Department (RID)
01000	8	*reserved*	State Emergency Service New South Wales (SES)	*reserved*	Department of Pollution Control (DPC)

Code [5 bits]	Provider ID	Provider Name (Japan)	Provider Name (Australia)	Provider Name (Fiji)	Provider Name (Thailand)
01001	9	*reserved*	New South Wales Rural Fire Service (NSW RFS)	*reserved*	Geo-Informatics and Space Technology Development Agency (GISTDA)
01010	10	*reserved*	Joint Australian Tsunami Warning Centre (JATWC)	*reserved*	Electricity Generating Authority of Thailand (EGAT)
01011	11	*reserved*	Flood Knowledge Centre (FKC)	*reserved*	Royal Forest Department (RFD)
01100	12	*reserved*	Australian Broadcasting Corporation (ABC)	*reserved*	Department of Parks, Wildlife and Plant Conservation (DPWPC)
01101	13	*reserved*	*reserved*	*reserved*	Water Crisis Prevention Center (WCPC)
01110	14	*reserved*	*reserved*	*reserved*	*reserved*
01111	15	*reserved*	*reserved*	*reserved*	*reserved*
10000	16	*reserved*	*reserved*	*reserved*	*reserved*
10001	17	*reserved*	*reserved*	*reserved*	*reserved*
10010	18	*reserved*	*reserved*	*reserved*	*reserved*
10011	19	*reserved*	*reserved*	*reserved*	*reserved*
10100	20	*reserved*	*reserved*	*reserved*	*reserved*
10101	21	*reserved*	*reserved*	*reserved*	*reserved*
10110	22	*reserved*	*reserved*	*reserved*	*reserved*
10111	23	*reserved*	*reserved*	*reserved*	*reserved*
11000	24	*reserved*	*reserved*	*reserved*	*reserved*
11001	25	*reserved*	*reserved*	*reserved*	*reserved*
11010	26	*reserved*	*reserved*	*reserved*	*reserved*
11011	27	*reserved*	*reserved*	*reserved*	*reserved*
11100	28	*reserved*	*reserved*	*reserved*	*reserved*
11101	29	*reserved*	*reserved*	*reserved*	*reserved*
11110	30	*reserved*	*reserved*	*reserved*	*reserved*
11111	31	*reserved*	*reserved*	*reserved*	*reserved*

4.2.3.4. A4-Hazard Category and Type

This field is a 7-bit code indicating the hazard type of the issued alert.

Table 4.2-7 shows the contents of Hazard Category and Type.

See the applicable document (2) for information about the details of the code.

Table 4.2-7 Hazard Category and Type

Code [7 bits]	Hazard Category and Type
0000000	*not used*
0000001	CBRNE - Air Strike
0000010	CBRNE - Attack on IT Systems
・・・	・・・
0010010	ENVIRONMENT - Marine Pollution
・・・	・・・
0011011	FIRE - Forest Fire
・・・	・・・
0101100	GEO - Tsunami
・・・	・・・
0110011	HEALTH - Pandemic
・・・	・・・
1001101	MET - Storm or Thunderstorm
・・・	・・・
1110000	TRANSPORT - Tunnel accident
1110001	OTHER - Test Alert
1110010	*reserved*
・・・	・・・
1111111	*reserved*

4.2.3.5. A5-Severity

This field is a 2-bit code indicating the severity of the alert.

Table 4.2-8 shows the contents of Severity.

Table 4.2-8 Severity

Code [2bits]	Severity
00	Unknown
01	Moderate-Possible threat to life or property
10	Severe - Significant threat to life or property
11	Extreme - Extraordinary threat to life or property

4.2.3.6. A6-Hazard Onset: Week Number

This field is a 1-bit code indicating the week a hazard is expected to occur (either the current week or the following week).

Table 4.2-9 shows the contents of Hazard Onset: Week Number.

Table 4.2-9 Hazard Onset: Week Number

Code [1bit]	Week
0	Current
1	Next

4.2.3.7. A7-Hazard Onset: Time of the Week

This field is a 14-bit code indicating the exact time a hazard is expected to occur within the week specified in the A6-Hazard Onset: Week Number, with a resolution of 1 minute. The time is represented in UTC.

Table 4.2-10 shows the contents of Hazard Onset: Time of the Week.

Table 4.2-10 Hazard Onset: Time of the Week

Code [14 bits]	WEEKDAY - Time [UTC]
00000000000000	*not used*
00000000000001	MONDAY - 00:00 AM
00000000000010	MONDAY - 00:01 AM
00000000000011	MONDAY - 00:02 AM
・・・	・・・
10011101011110	SUNDAY - 11:57 PM
10011101011111	SUNDAY - 11:58 PM
10011101100000	SUNDAY - 11:59 PM
10011101100001	*reserved*
・・・	・・・
11111111111111	*reserved*

4.2.3.8. A8-Hazard Duration

This field is a 2-bit code indicating the valid duration in which an alert is expected (the unit is the hour (H)).

Table 4.2-11 shows the contents of Hazard Duration.

This field can be used for the recipient to clear expired alerts. Even if the "All Clear" message is not transmitted, the receiving side can judge whether an alert is active or not, by referencing this field and comparing its value with the current time.

Table 4.2-11 Hazard Duration

Code [2bits]	Duration
00	Unknown
01	Duration < 6H
10	6H <= Duration < 12H
11	12H <= Duration < 24H

4.2.3.9. A9-Type of Library

This field uses a 1-bit code to distinguish between the International library and the Country/region library for identifying which type of library to utilize.

Table 4.2-12 shows the contents of Type of Library.

The International library was created by the European Commission based on the inputs from EU member nations.

The Country/region library is one created by each individual country or region. In the case of DCX message (L-Alert), DCX message (J-Alert), and DCX message (information from local government), the code is fixed to "1" (Country/region library).

Table 4.2-12 Type of Library

Code [1 bit]	Library
0	International library
1	Country/region library

4.2.3.10. A10-Version of Library

This field is a 3-bit code indicating the version of the library to be used.

Table 4.2-13 shows the contents of Version of Library.

In the case of DCX message (L-Alert), DCX message (J-Alert), and DCX message (information from local government), the code is fixed to "000" (Library version = #1).

Table 4.2-13 Version of Library

Code [3 bits]	Library version
000	#1
001	#2
010	#3
011	#4
100	#5
101	#6
110	#7
111	#8

4.2.3.11. A11-Instruction library

This field is a 10-bit code indicating evacuation action instructions.

The field consists of two different lists. An instruction is selected from each list, and the selected instructions are combined and used.

The contents of the lists are described below.

(1) DCX message (L-Alert), DCX message (J-Alert), and DCX message (information from local government)

Note: Because the field format and content differ from CAMF, refer to this section instead of applicable document (2).

The field consists of the following two lists.

・Basic instruction

・Basic instruction information

Figure 4.2-3 shows the format of the Instruction library field. Table 4.2-14 shows the contents of the field. The first two bits of the 10-bit field represent the "Basic Instruction," while the remaining eight bits represent the "Basic Instruction Information." When the first two bits are not 00, the evacuation action instructions are expressed by combining the two lists. When the first two bits are 00, the evacuation action instructions are expressed using the entire 10-bit value, rather than by combining the two lists. All patterns of evacuation action instructions are shown in Table 4.2-15.

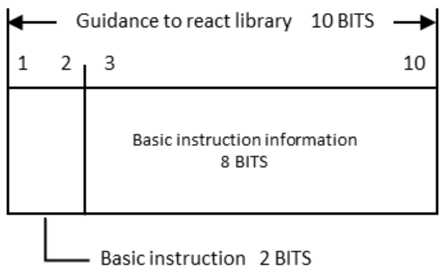


Figure 4.2-3 Format of the Instruction library Field (DCX message (L- Alert), DCX message (J-Alert), and DCX message (information from local government)

Table 4.2-14 Instruction library (1/2) Bits 1-2 (Basic instruction)

Code [bits 1-2]	Basic instruction
00	"" (Not specified)
01	"Stay"
10	"Move to/toward"
11	"Keep away from"

Table 4.2-14 Instruction library (2/2) Bits 3-10 (Basic instruction information)

Code [bits 3-10]	Basic instruction information
00000000	“” (Not specified)
00000001	“Under/inside a solid structure”
00000010	“3rd floor or higher”
00000011	“Underground”
00000100	“Mountain”
00000101	“Water area”
00000110	“Building where chemicals are handled, such as a factory”
00000111	“Cliffs and areas at risk of collapse”
00001000 to 01111110	*reserved*
01111111	“Take the best immediate action to save your life.”
10000000 to 11111110	Reserved for the evacuation action instruction text for J-Alert
11111111	“Take the best immediate action to save your life.”

Table 4.2-15 Evacuation Action Instruction List

A11 - Instruction Library[10 bits]		Evacuation Action Instruction Text
bit1-2	bit3-10	
"00"	"00000000"	"" (Not specified) In the case of DCX message (information from local government) , reference EX2 to EX7.
"00"	"00000001"	"Take the best immediate action to save your life." "直ちに命を守るための最善の行動を。"
"00"	"00000010" to "01111101"	*reserved*
"00"	"01111110"	"This is a test message for DCX." "これは、DCX のテストです。"
"00"	"01111111"	"Take the best immediate action to save your life." "直ちに命を守るための最善の行動を。"
"00"	"10000000"	"Missile launched, missile launched. It is believed that a missile was launched. Please take shelter inside buildings or underground." "ミサイル発射。ミサイル発射。ミサイルが発射されたものとみられます。建物の中、又は地下に避難して下さい。"
"00"	"10000001"	"Missile passed, missile passed. It is believed that the previous missile has passed over the area. The call for evacuation will be canceled. If you find any suspicious object, please stay away from it and inform the police or the fire department immediately." "ミサイル通過。ミサイル通過。先程のミサイルは通過したものとみられます。避難の呼びかけを解除します。不審なものには決して近寄らず直ちに警察や消防などに連絡して下さい。"

A11 - Instruction Library[10 bits]		Evacuation Action Instruction Text
bit1-2	bit3-10	
"00"	"10000010"	"It is believed that the previous missile has dropped in the sea. The call for evacuation will be canceled. If you find any suspicious object, please stay away from it and inform the police or the fire department immediately."
		"先程のミサイルは、海に落下したものとみられます。避難の呼びかけを解除します。不審なものには決して近寄らず直ちに警察や消防などに連絡して下さい。"
"00"	"10000011"	"It is believed that the previous missile will not come to Japan. The call for evacuation will be canceled."
		"先程のミサイルは、我が国には飛来しないものとみられます。避難の呼びかけを解除します。"
"00"	"10000100"	"Take shelter immediately, take shelter immediately. Please take shelter inside buildings or underground. It is believed that a missile will drop around this area. Please take shelter immediately."
		"直ちに避難。直ちに避難。直ちに建物の中、又は地下に避難して下さい。ミサイルが、周辺に落下するものとみられます。直ちに避難して下さい。"
"00"	"10000101"	"The previous missile has been intercepted and destroyed. There is a possibility of pieces of the destroyed missile dropping. We will keep you informed. Please stay indoors for shelter."
		"先程のミサイルは、迎撃により破壊されました。ミサイルの破片の落下の可能性があります。続報を伝達しますので、引き続き屋内に避難して下さい。"
"00"	"10000110"	"Missile dropped, missile dropped. It is believed that a missile has dropped around this area. We will keep you informed. Please stay indoors for shelter."
		"ミサイル落下。ミサイル落下。ミサイルが、周辺に落下したものとみられます。続報を伝達しますので、引き続き屋内に避難して下さい。"
"00"	"10000111"	"It is believed that the previous missile will not drop in Japan. The call for evacuation will be canceled."
		"先程のミサイルは、我が国には落下しないものとみられます。避難の呼びかけを解除します。"
"00"	"10001000"	"This is a test message for J-Alert."
		"これは、J アラートのテストです。"
"00"	"10001001" to "11111110"	To be used in the evacuation action instruction text for J-Alert (*reserved*).

A11 - Instruction Library[10 bits]		Evacuation Action Instruction Text
bit1-2	bit3-10	
"00"	"11111111"	"Take immediate action to save your life." "直ちに命を守るための最善の行動を。"
"01"	"00000000"	"Stay." "留まれ。"
"01"	"00000001"	"Stay. Under/inside a solid structure." "留まれ。頑丈なものの下/中。"
"01"	"00000010"	"Stay. 3rd floor or higher." "留まれ。3 階以上。"
"01"	"00000011"	"Stay. Underground." "留まれ。地下。"
"01"	"00000100"	"Stay. Mountain." "留まれ。山。"
"01"	"00000101"	"Stay. Water area." "留まれ。水場。"
"01"	"00000110"	"Stay. Building where chemicals are handled, such as a factory." "留まれ。工場等化学系を取扱う建物。"
"01"	"00000111"	"Stay. Cliffs and areas at risk of collapse." "留まれ。崖等崩れやすい場所。"
"01"	"00001000" to "01111111"	*reserved*
"01"	"10000000" to "11111111"	To be used in the evacuation action instruction text for J-Alert (reserved).
"10"	"00000000"	" Move to/toward" "向かえ。"
"10"	"00000001"	"Move to/toward Under/inside a solid structure." "向かえ。頑丈なものの下/中。"
"10"	"00000010"	"Move to/toward 3rd floor or higher." "向かえ。3 階以上。"
"10"	"00000011"	"Move to/toward Underground." "向かえ。地下。"
"10"	"00000100"	"Move to/toward Mountain." "向かえ。山。"
"10"	"00000101"	"Move to/toward Water area." "向かえ。水場。"
"10"	"00000110"	"Move to/toward Building where chemicals are handled, such as a factory." "向かえ。工場等化学系を取扱う建物。"
"10"	"00000111"	"Move to/toward Cliffs and areas at risk of collapse." "向かえ。崖等崩れやすい場所。"
"10"	"00001000" to "01111111"	*reserved*
"10"	"10000000" to "11111111"	Reserved for the evacuation action instruction text for J-Alert.
"11"	"00000000"	"Keep away from" "離れろ。"
"11"	"00000001"	"Keep away from Under/inside a solid structure." "離れろ。頑丈なものの下/中。"

A11 - Instruction Library[10 bits]		Evacuation Action Instruction Text
bit1-2	bit3-10	
"11"	"00000010"	"Keep away from 3rd floor or higher."
		"離れろ。3 階以上。"
"11"	"00000011"	"Keep away from Underground."
		"離れろ。地下。"
"11"	"00000100"	"Keep away from Mountain."
		"離れろ。山。"
"11"	"00000101"	"Keep away from Water area."
		"離れろ。水場。"
"11"	"00000110"	"Keep away from Building where chemicals are handled, such as a factory."
		"離れろ。工場等化学系を取扱う建物。"
"11"	"00000111"	"Keep away from Cliffs and areas at risk of collapse."
		"離れろ。崖等崩れやすい場所。"
"11"	"00001000" to "01111111"	*reserved*
"11"	"10000000" to "11111111"	Reserved for the evacuation action instruction text for J-Alert.

(2) DCX message (information from organizations outside Japan)

When bit 1 of the A9 field is “0” (International library), the field consists of the two lists mentioned below.

・List A: General required action

・List B: Monitoring + Required action

Figure 4.2-4 shows the format of the field. Table 4.2-16 shows the contents of the field.

When bit 1 of the A9 field is “1” (Country/region library), the library created by the country or region indicated in the A2 field is used. Details are undecided.

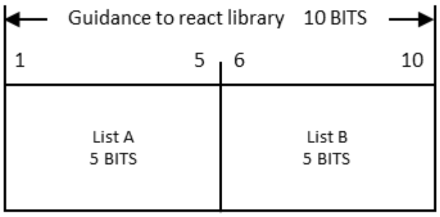


Figure 4.2-4 Format of the Instruction library Field (DCX message (information from organizations outside Japan))

Table 4.2-16 Instruction library (1/2) Bits 1-5 (List A: General required action)

Code [bits 1-5]	Instruction Code	Instruction list – EN
00000	IC-A-01	[*reserved*]
00001	IC-A-02	You are in the danger zone, leave the area immediately. Listen to radio or media for directions and information.
00010	IC-A-03	You are in the danger zone, leave the area immediately and reach the evacuation point indicated by the area plotted in yellow. Listen to radio or media for directions and information.
00011	IC-A-04	Seek shelter in a building immediately. Stay under cover and stay informed.
・・・	・・・	・・・
11111	IC-A-32	Conditions have improved and are no longer expected to meet alert criteria.

Table 4.2-16 Instruction library (2/2) Bits 6-10 (List B: Monitoring + Required action)

Code [bits 6-10]	Instruction Code	Instruction list – EN
00000	IC-B-01	[*reserved*]
00001	IC-B-02	Check with the weather services and local authorities for additional information
00010	IC-B-03	Find out the location of the information points set up by the authorities on official channels (radio, internet, TV, social networks…)
00011	IC-B-04	Sensitive or vulnerable people should not go out unless they must.
・・・	・・・	・・・
11111	IC-B-32	This replaces the warning previously in effect for this area.

4.2.3.12. A12-Ellipse Centre Latitude

This field is a 16-bit code indicating the ellipse centre latitude.

Latitude in the range of -90° to 90° is expressed while equally divided by the assigned 16 bits.

The relational expression of the code value and latitude is shown below. (A12 is converted to a decimal number when assigned.)

$$\text{Latitude } [deg] = -90 + \frac{180}{2^{16} - 1} A12$$

(Calculation example) When A12 = “1011001011000001”

When “1011001011000001” is converted to a decimal number, the result is “45761”.

This is assigned to the above expression. The number of decimal places shown for the calculation result is 3 (rounded off at the fourth decimal place). Therefore, the calculation is as follows.

Latitude [deg] = -90＋(180÷((2^16)-1))×45761≒35.688258

For DCX message (J-Alert), this field is not used (fixed to “0000000000000000”).

4.2.3.13. A13-Ellipse Centre Longitude

This field is a 17-bit code indicating the ellipse centre longitude.

Longitude in the range of -180° to 180° is expressed while equally divided by the assigned 17 bits.

The relational expression of the code value and longitude is shown below. (A13 is converted to a decimal number when assigned.)

$$\text{Longitude } [deg] = -180 + \frac{360}{2^{17} - 1} A13$$

(Calculation example) When A13 = “11100011010101011”

When “11100011010101011” is converted to a decimal number, the result is “116395”.

This is assigned to the above expression. The number of decimal places shown for the calculation result is 3 (rounded off at the fourth decimal place). Therefore, the calculation is as follows.

Longitude [deg] = -180＋(360÷((2^17)-1))×116395≒139.690855

For DCX message (J-Alert), this field is not used (fixed to “00000000000000000”).

4.2.3.14. A14-Ellipse Semi-Major Axis

This field is a 5-bit code indicating the ellipse semi-major axis length.

The relational expression of the code value and ellipse semi-major axis length is shown below. (a is an integer between 0 and 31.)

$$Radius[m] = 10^{Log10(Min\ Radius)+\frac{a\ (Log10(Max\ Radius)-Log10(Min\ Radius))}{Max\ a}}$$

Max Radius[km] =2500

Min Radius[km] =0.216

Max a =31

The number of decimal places shown for the calculation result is 3 (rounded off at the fourth decimal place).

Table 4.2-17 shows the contents of Ellipse Semi-Major Axis.

Table 4.2-17 Ellipse Semi-Major Axis

Code [5 bits]	a	Radius [km]
00000	0	0.216
00001	1	0.292
00010	2	0.395
00011	3	0.535
00100	4	0.723
00101	5	0.978
00110	6	1.322
00111	7	1.788
01000	8	2.418
01001	9	3.269
01010	10	4.421
01011	11	5.979
01100	12	8.085
01101	13	10.933
01110	14	14.784
01111	15	19.992
10000	16	27.035
10001	17	36.559
10010	18	49.439
10011	19	66.855
10100	20	90.407
10101	21	122.255
10110	22	165.324
10111	23	223.564
11000	24	302.322
11001	25	408.824
11010	26	552.846
11011	27	747.603
11100	28	1010.970
11101	29	1367.116

Code [5 bits]	a	Radius [km]
11110	30	1848.727
11111	31	2500.000

For DCX message (J-Alert), this field is not used (fixed to “00000”).

4.2.3.15. A15-Ellipse Semi-Minor Axis

This field is a 5-bit code indicating the ellipse semi-minor axis length.

The relational expression of the code value and ellipse semi-minor axis length is shown below. (a is an integer between 0 and 31.)

$$Radius[m] = 10^{Log10(Min\ Radius)+\frac{a\ (Log10(Max\ Radius)-Log10(Min\ Radius))}{Max\ a}}$$

Max Radius[km] =2500

Min Radius[km] =0.216

Max a =31

The number of decimal places shown for the calculation result is 3 (rounded off at the fourth decimal place).

Table 4.2-18 shows the contents of Ellipse Semi-Minor Axis.

Table 4.2-18 Ellipse Semi-Minor Axis

Code [5 bits]	a	Radius [km]
00000	0	0.216
00001	1	0.292
00010	2	0.395
00011	3	0.535
00100	4	0.723
00101	5	0.978
00110	6	1.322
00111	7	1.788
01000	8	2.418
01001	9	3.269
01010	10	4.421
01011	11	5.979
01100	12	8.085
01101	13	10.933
01110	14	14.784
01111	15	19.992
10000	16	27.035
10001	17	36.559
10010	18	49.439
10011	19	66.855
10100	20	90.407
10101	21	122.255
10110	22	165.324
10111	23	223.564
11000	24	302.322
11001	25	408.824
11010	26	552.846
11011	27	747.603
11100	28	1010.970

Code [5 bits]	a	Radius [km]
11101	29	1367.116
11110	30	1848.727
11111	31	2500.000

For DCX message (J-Alert), this field is not used (fixed to “00000”).

4.2.3.16. A16-Ellipse Azimuth

This field is a 6-bit code indicating the ellipse azimuth angle.

The azimuth angle in the range of -90° to 90° is expressed while equally divided by the assigned 6 bits. The azimuth angle of the ellipse is measured from East to the ellipse major axis line, with positive values counted from East toward North.

The relational expression of the code value and azimuth angle is shown below. (A16 is converted to a decimal number when assigned.)

$$\text{Azimuth angle}[deg] = -90 + \frac{180}{2^6} A16$$

(Calculation example) When A16 = “110000”

When “110000” is converted to a decimal number, the result is “48”.

This is assigned to the above expression. The number of decimal places shown for the calculation result is 2 (rounded off at the third decimal place). Therefore, the calculation is as follows.

Azimuth angle [deg] = -90＋(180÷2^6)×48 = 45.00

For DCX message (J-Alert), this field is not used (fixed to “000000”).

4.2.3.17. A17-Type of Specific Settings

This field is a 2-bit code indicating the category of additional information regarding the hazard. Table 4.2-19 shows the contents of Type of Specific Settings.

For DCX message (L-Alert) and DCX message (information from local government), the category is fixed to “00” (B1 - Improved Resolution of Main Ellipse).

For DCX message (J-Alert), this field is not used. (Fixed to “00”)

Table 4.2-19 Type of Specific Settings

Code [bits 1-2]	Category	Comments
00	B1 - Improved Resolution of Main Ellipse	The parameters improve the resolution of the center of the ellipse and of its axes.
01	B2 - Position of the Centre of the Hazard	When centre of hazard is different from the centre of the ellipse. It is expressed in degrees of latitude and longitude in offset of the center of the ellipse, with a limitation of +/- 10 degrees.
10	B3 - Secondary Ellipse Definition	Geometry of this second ellipse is computed from the original ellipse (shift, size and rotation).
11	B4 - Quantitative and detailed information about the Hazard	Option to provide quantitative or more detailed information related to the hazard category and type.

4.2.3.18. A18-Specific Settings

This field is a 15-bit code indicating additional information regarding the hazard.

See the applicable document (2) for details.

For DCX message (J-Alert), this field is not used (fixed to”000000000000000”)

4.2.4. Extended Message

This message is a 74-bit extended area containing DCX message-specific information. The contents of each DCX message are described below.

4.2.4.1. DCX message (L-Alert), DCX message (information from local government)

Figure 4.2-5 shows the message format. Table 4.2-20 shows the parameter definition.

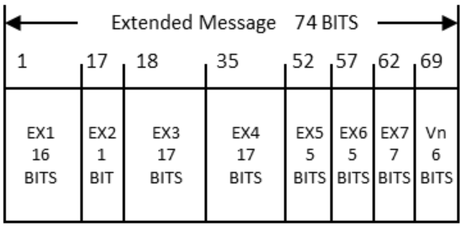


Figure 4.2-5 Extended Message Format (DCX message (L-Alert) and DCX message (information from local government)

Table 4.2-20 Parameter Definition (DCX message (L-Alert) and DCX message (information from local government))

Parameter	Description	Number of Bits
EX1	Target Area Code → See 4.2.4.1.1	16
EX2	Evacuate Direction Type → See 4.2.4.1.2	1
EX3	Additional Ellipse Centre Latitude → See 4.2.4.1.3	17
EX4	Additional Ellipse Centre Longitude → See 4.2.4.1.4	17
EX5	Additional Ellipse Semi-Major Axis → See 4.2.4.1.5	5
EX6	Additional Ellipse Semi-Minor Axis → See 4.2.4.1.6	5
EX7	Additional Ellipse Azimuth → See 4.2.4.1.7	7
Vn	Version number → See 4.2.4.1.8	6

4.2.4.1.1. EX1-Target Area Code

This field is a 16-bit code indicating the alert target city, town and village.

A value coded based on the applicable document (3) is set.

Table 4.2-21 shows the contents of Target Area Code .

Coding is performed by removing the least significant digit (check digit) from the 6-digit code of applicable document (3).

(Coding example)

The code of Sapporo City of Hokkaido is "011002" (in decimal notation).

The least significant digit is removed, and the value converted to a 16-bit binary number is set in EX1.

"1100" (decimal notation) → "0000010001001100" (binary notation)

If the target area is not specified, ALL0 is set.

This field is used when the ellipse-based target area judgment is not made (A12 to A16 fields are all set to "0").

Table 4.2-21 Target Area Codes

Code [16 bits]	Prefecture Name	City, Town and Village Name
0000010001001100	Hokkaido	Sapporo
0000010010110010	Hokkaido	Hakodate
0000010010110011	Hokkaido	Otaru
・・・	・・・	・・・
1011100100010101	Okinawa	Taketomi
1011100100010110	Okinawa	Yonaguni

4.2.4.1.2. EX2-Evacuate Direction Type

This field is a 1-bit code indicating the action instruction for the additional target area.

Table 4.2-22 shows the contents of Evacuate Direction Type.

The additional target area is that enclosed in the ellipse indicated by the EX3 to EX7 fields.

This field is used for DCX message (information from local government).

This field is not used for DCX message (L-Alert).

Table 4.2-22 EX2-Evacuate Direction Type

Code [bit 1]	Instruction
0	"Leave the additional target area range."
1	"Head to the additional target area range."

4.2.4.1.3. EX3-Additional Ellipse Centre Latitude

This field is a 17-bit code indicating the ellipse centre latitude of the additional target area.

Latitude in the range of -90° to 90° is expressed while equally divided by the assigned 17 bits.

The relational expression of the code value and latitude is shown below. (EX3 is converted to a decimal number when assigned.)

$$\text{Latitude } [deg] = -90 + \frac{180}{2^{17} - 1} EX3$$

(Calculation example) When EX3 = "10110010110000010"

When "10110010110000010" is converted to a decimal number, the result is "91522".

This is assigned to the above expression. The number of decimal places shown for the calculation result is 3 (rounded off at the fourth decimal place). Therefore, the calculation is as follows.

Latitude [deg] = -90＋(180÷((2^17)-1))×91522≒35.687299

This field is used for DCX message (information from local government).

This field is not used for DCX message (L-Alert).

4.2.4.1.4. EX4-Additional Ellipse Centre Longitude

This field is a 17-bit code indicating the ellipse centre longitude of the additional target area.

Longitude in the range of 45° to 225° is expressed while equally divided by the assigned 17 bits.

The relational expression of the code value and longitude is shown below. (EX4 is converted to a decimal number when assigned.)

$$\text{Longitude } [deg] = 45 + \frac{180}{2^{17} - 1} EX4$$

* It is sufficient that a range of 180° can be expressed with the center being the orbiting range of the QZSS. Therefore, the range is 45° to 225° with the center being 135°.

(Calculation example) When EX4 = "10000110101010110"

When "10000110101010110" is converted to a decimal number, the result is "68950".

This is assigned to the above expression. The number of decimal places shown for the calculation result is 3 (rounded off at the fourth decimal place). Therefore, the calculation is as follows.

Longitude [deg] = 45＋(180÷((2^17)-1))×68950≒139.689138

This field is used for DCX message (information from local government).

This field is not used for DCX message (L-Alert).

4.2.4.1.5. EX5-Additional Ellipse Semi-Major Axis

This field is a 5-bit code indicating the ellipse semi-major axis length of the additional target area.

See Table 4.2-17.

This field is used for DCX message (information from local government).

This field is not used for DCX message (L-Alert).

4.2.4.1.6. EX6-Additional Ellipse Semi-Minor Axis

This field is a 5-bit code indicating the ellipse semi-minor axis length of the additional target area.

See Table 4.2-18.

This field is used for DCX message (information from local government).

This field is not used for DCX message (L-Alert).

4.2.4.1.7. EX7-Additional Ellipse Azimuth

This field is a 7-bit code indicating the ellipse azimuth angle of the additional target area.

The azimuth angle in the range of -90° to 90° is expressed while equally divided by the assigned 7 bits. The azimuth angle of the ellipse is measured from East to the ellipse major axis line, with positive values counted from East toward North.

The relational expression of the code value and azimuth angle is shown below. (EX7 is converted to a decimal number when assigned.)

$$\text{Azimuth angle}[deg] = -90 + \frac{180}{2^7}EX7$$

(Calculation example) When EX7 = "1100000"

When "1100000" is converted to a decimal number, the result is "96".

This is assigned to the above expression. The number of decimal places shown for the calculation result is 2 (rounded off at the third decimal place). Therefore, the calculation is as follows.

Azimuth angle [deg] = -90＋(180÷2^7)×96 = 45.00

This field is used for DCX message (information from local government).

This field is not used for DCX message (L-Alert).

4.2.4.1.8. Vn-Version Number

This field is a 6-bit code indicating the version number of DCX message (L-Alert) or DCX message (information from local government). The value is fixed to "000001".

4.2.4.2. DCX message (J-Alert)

Figure 4.2-6 shows the message format. Table 4.2-23 shows the parameter definition.

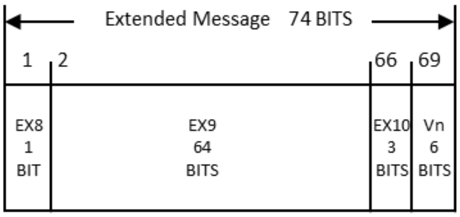


Figure 4.2-6 Extended Message Format (DCX message (J-Alert))

Table 4.2-23 Parameter Definition (DCX Message (J-Alert))

Parameter	Description	Number of Bits
EX8	Target Area Code List Type → See 4.2.4.2.1	1
EX9	Target Area Code List → See 4.2.4.2.2	64
EX10	*reserved*	3
Vn	Version number → See 4.2.4.2.3	6

4.2.4.2.1. EX8-Target Area Code List Type

This field is a 1-bit code indicating whether the contents of the EX9-Target Area Code List are prefecture codes or cities, towns and villages codes.

Table 4.2-24 shows the contents of the Target Area Code List Type.

Table 4.2-24 EX8-Target Area Code List Type

Code [bit 1]	Target Area Code List Type
0	Prefecture code
1	Cities, towns and villages code

4.2.4.2.2. EX9-Target Area Code List

This field is a 64-bit code indicating the alert target prefecture or cities, towns and villages. Whether the field indicates a prefecture or cities, towns and villages is determined by the contents of EX8.

(1) In the case of a prefecture

The field consists of a 47-bit prefecture code and 17 reserved bits.
Figure 4.2-7 shows the field format.

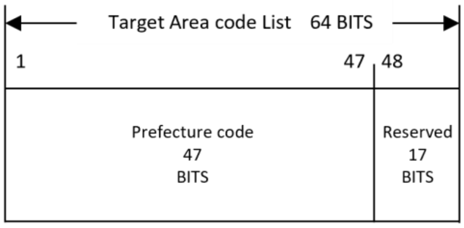


Figure 4.2-7 Format of the Target Area Code List Field (Prefecture)

Table 4.2-25 shows the prefecture codes.

When the OR operation result of multiple codes is set, multiple prefectures are the target.

For example, in the case of "00000000000000000000000000000000000000000000111", Hokkaido, Aomori, and Iwate are the target area.

When all the bits are set to "1", all the prefectures are the target area.

Table 4.2-25 Prefecture Codes

Code [47 bits]	Prefecture Name	Effective (= "1") Bit Position
00000000000000000000000000000000000000000000001	Hokkaido	47
00000000000000000000000000000000000000000000010	Aomori	46
00000000000000000000000000000000000000000000100	Iwate	45
00000000000000000000000000000000000000000001000	Miyagi	44
00000000000000000000000000000000000000000010000	Akita	43
00000000000000000000000000000000000000000100000	Yamagata	42
00000000000000000000000000000000000000001000000	Fukushima	41
00000000000000000000000000000000000000010000000	Ibaraki	40
00000000000000000000000000000000000000100000000	Tochigi	39
00000000000000000000000000000000000001000000000	Gunma	38

Code [47 bits]	Prefecture Name	Effective (= "1") Bit Position
00000000000000000000000000000000000010000000000	Saitama	37
00000000000000000000000000000000000100000000000	Chiba	36
00000000000000000000000000000000001000000000000	Tokyo	35
00000000000000000000000000000000010000000000000	Kanagawa	34
00000000000000000000000000000000100000000000000	Niigata	33
00000000000000000000000000000001000000000000000	Toyama	32
00000000000000000000000000000010000000000000000	Ishikawa	31
00000000000000000000000000000100000000000000000	Fukui	30
00000000000000000000000000001000000000000000000	Yamanashi	29
00000000000000000000000000010000000000000000000	Nagano	28
00000000000000000000000000100000000000000000000	Gifu	27
00000000000000000000000001000000000000000000000	Shizuoka	26
00000000000000000000000010000000000000000000000	Aichi	25
00000000000000000000000100000000000000000000000	Mie	24
00000000000000000000001000000000000000000000000	Shiga	23
00000000000000000000010000000000000000000000000	Kyoto	22
00000000000000000000100000000000000000000000000	Osaka	21
00000000000000000001000000000000000000000000000	Hyogo	20
00000000000000000010000000000000000000000000000	Nara	19
00000000000000000100000000000000000000000000000	Wakayama	18
00000000000000001000000000000000000000000000000	Tottori	17
00000000000000010000000000000000000000000000000	Shimane	16
00000000000000100000000000000000000000000000000	Okayama	15
00000000000001000000000000000000000000000000000	Hiroshima	14
00000000000010000000000000000000000000000000000	Yamaguchi	13
00000000000100000000000000000000000000000000000	Tokushima	12
00000000001000000000000000000000000000000000000	Kagawa	11
00000000010000000000000000000000000000000000000	Ehime	10
00000000100000000000000000000000000000000000000	Kochi	9
00000001000000000000000000000000000000000000000	Fukuoka	8
00000010000000000000000000000000000000000000000	Saga	7
00000100000000000000000000000000000000000000000	Nagasaki	6
00001000000000000000000000000000000000000000000	Kumamoto	5
00010000000000000000000000000000000000000000000	Oita	4

Code [47 bits]	Prefecture Name	Effective (= "1") Bit Position
00100000000000000000000000000000000000000000000	Miyazaki	3
01000000000000000000000000000000000000000000000	Kagoshima	2
10000000000000000000000000000000000000000000000	Okinawa	1

(2) In the case of a cities, towns and villages

The field consists of four 16-bit cities, towns and villages codes. (Up to 4 codes are set.) Figure 4.2-8 shows the field format.

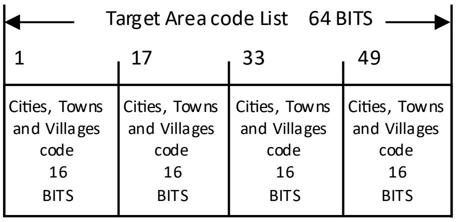


Figure 4.2-8 Format of the Target Area Code List Field (Cities, Towns and Villages)

See 4.2.4.1.1for information about the cities, towns and villages codes.

When no cities, towns and villages are specified, ALL0 is set.

4.2.4.2.3. Vn-Version Number

This field is a 6-bit code indicating the version number of DCX message (J-Alert). The value is fixed to "000001".

4.2.4.3. DCX message (information from organizations outside Japan)

Figure 4.2-9 shows the message format. Table 4.2-26 shows the parameter definition.

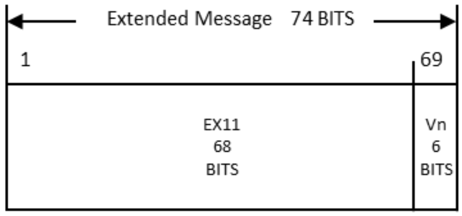


Figure 4.2-9 Extended Message Format (DCX message (information from organizations outside Japan))

Table 4.2-26 Parameter Definition (DCX message (information from organizations outside Japan))

Parameter	Description	Number of Bits
EX11	Organization outside Japan-Specific Area → See 4.2.4.3.1	68
Vn	Version Number → See 4.2.4.3.2	6

4.2.4.3.1. EX11-Organization Outside Japan-Specific Area

This field is a 68-bit code indicating organization outside Japan-specific information. Details are undefined.

4.2.4.3.2. Vn-Version Number

This field is defined under the unique version management system of an organization outside Japan.

4.3. MT44 NULL Message

This message is to transmit the Satellite Designation (SD field) to the receiver even when there is no information to transmit a DCX message. Since the message is dedicated to the transmission of the satellite designation information, it is not processed as a DCX message.

For the values to be set in each field of the message, see Table 4.3-1.

Table 4.3-1 MT44 NULL Message

Message Field		Number of Bits	Setting Value
PAB		8	—
MT		6	"101100" (MT44)
SD		10	See 4.2.2
Common Alert Message Format		—	—
	A1-Message Type	2	All bits set to "0"
	A2-Country/Region Name	9	"001101111" (Japan) *1
	A3-Provider Identifier	5	All bits set to "0"
	A4-Hazard Category and Type	7	All bits set to "0"
	A5-Severity	2	All bits set to "0"
	A6-Hazard Onset：Week Number	1	”0”
	A7-Hazard Onset：Time of Week	14	All bits set to "0"
	A8-Hazard Duration	2	All bits set to "0"
	A9-Type of Library	1	”0”
	A10-Version of Library	3	All bits set to "0"
	A11-Instruction library	10	All bits set to "0"
	A12-Ellipse Centre Latitude	16	All bits set to "0"
	A13-Ellipse Centre Longitude	17	All bits set to "0"
	A14-Ellipse Semi-Major Axis	5	All bits set to "0"
	A15-Ellipse Semi-Minor Axis	5	All bits set to "0"
	A16-Ellipse azimuth	6	All bits set to "0"
	A17-Type of Specific Settings	2	All bits set to "0"
	A18- Specific Settings	15	All bits set to "0"
Extended Message		74	All bits set to "0"
Reserved		6	—
CRC		24	—

*1 "001101111" (Japan) is also transmitted from the satellites allocated to organizations outside Japan.

4.4. DCX Message Output from the Receiver

This section describes how to output a DCX message from the receiver. The sentence format shown below is recommended for DCX message output from the receiver.

4.4.1. Sentence Format

- Output all characters in ASCII code.
- Start every sentence with the message header "$QZQSM".
- Of the 8 bits representing the PRN number of L1S in binary notation, use the low-order 6 bits written in decimal notation to express the satellite number.
- The total length is 252 bits because 2 bits of "00" are added after the 250 bits of the DCX message.
- At the end of the sentence, add the carriage return <CR> or linefeed <LF> code.

Table 4.4-1 shows the sentence format.

Table 4.4-1 Sentence Format

Field	Value	Number of Characters
Message header	$QZQSM	6
Field delimiter	,	1
Satellite number	56,57,58,61(PRN184,185,186,189)	2
Field delimiter	,	1
DCX message		63
Field delimiter	*	1
Checksum		2

Example: $QZQSM, 56, 53B0604DE・・・・・・・9ABCDEFC*1F

5. User Algorithm

5.1. Relationship between Receiver Output and Notification

Figure 5.1-1 shows an image (example) of the relationship between receiver output and notification.

After receiving the L1S signal, the receiver outputs a message in the sentence format described in 4.4.1

Based on the receiver output, the CPU app references the DCX message and outputs it to the screen or other destination.

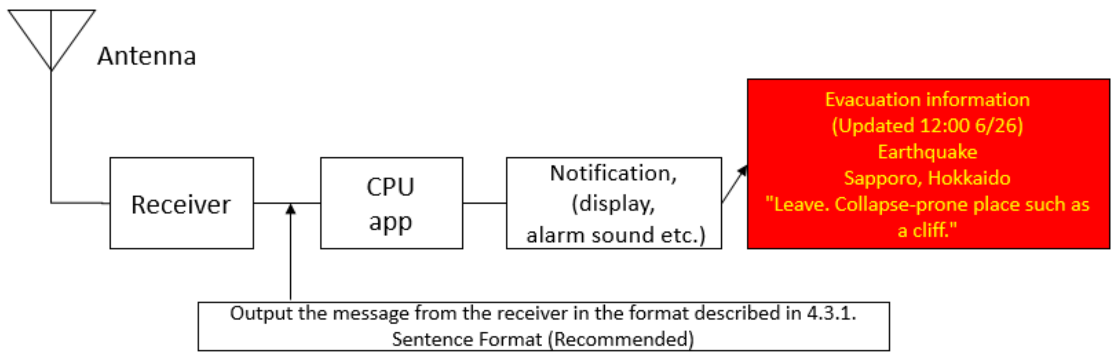


Figure 5.1-1 Image (Example) of the Relationship between Receiver Output and Notification

5.2. Service Area

DCX messages for Japan are available within the area shown in Figure 5.2-1, and DCX messages for use outside Japan are available within the area shown in Figure 5.2-2. The area in Figure 5.2-2 is where at least one QZS transmitting DCX messages for use outside Japan is visible with an elevation angle of 10 degrees or more. The actual area where a transmitted message is valid depends on the content of the message. See 4.2 for details.

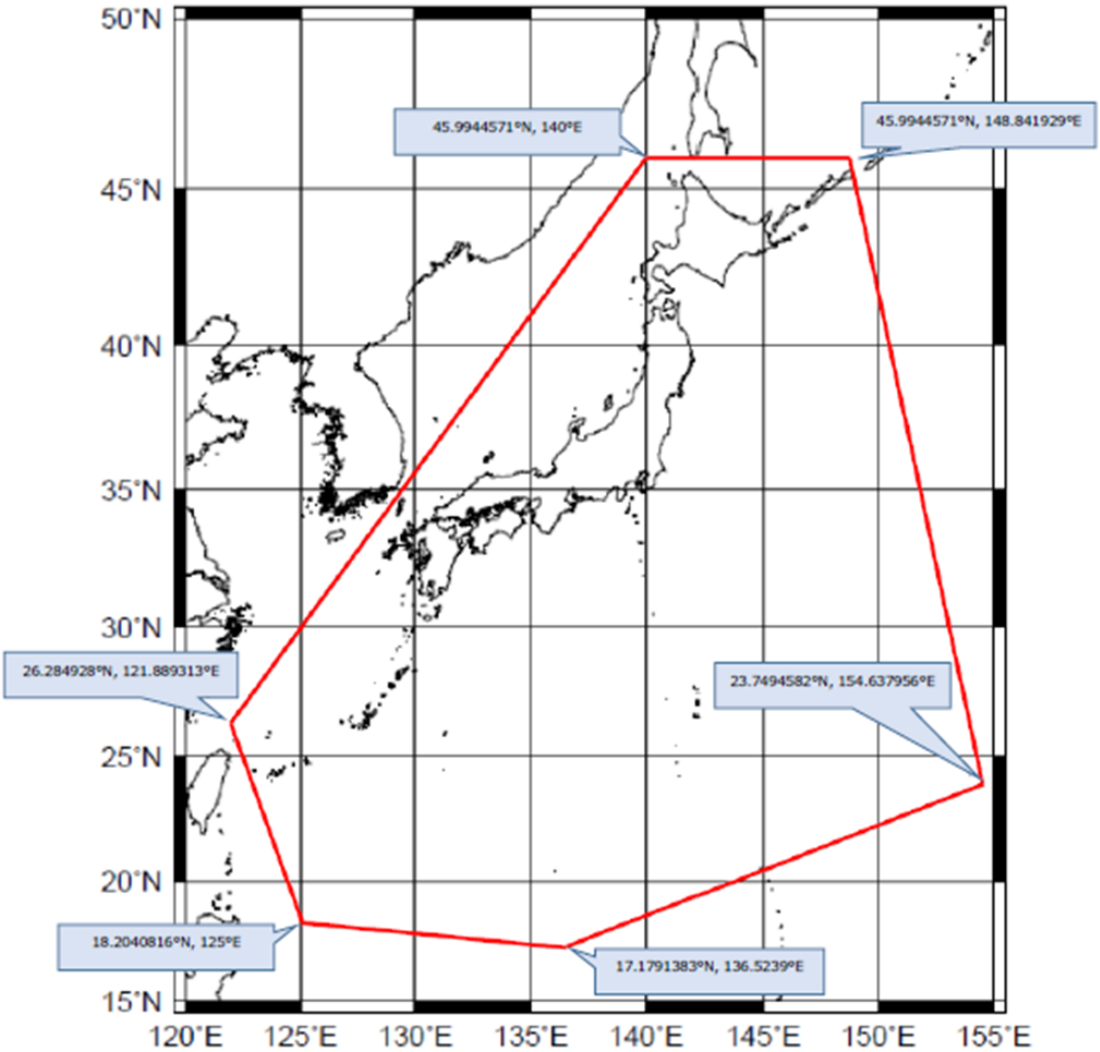


Figure 5.2-1　Service Area (DCX message (L-Alert), DCX message (J-Alert), and DCX message (information from local government))

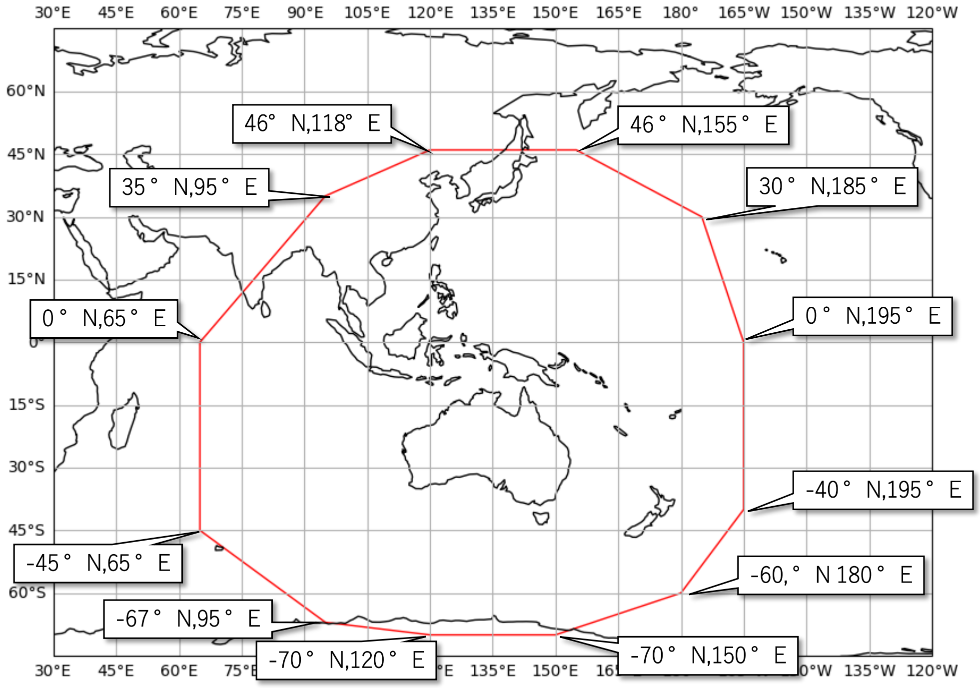


Figure 5.2-2 Service Area (DCX message (information from organizations outside Japan))

5.3. Examples of the Transmitted DCX message

5.3.1. DCX message (L-Alert)/ DCX message (information from local government)

(1) DCX message (L-Alert)

Format	PAB	MT	SD	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11
BIN	01010011	101100	000110000	01	001101111	00001	1001010	10	0	10010011001101	10	1	000	1100000101
HEX	5 3	B	0 6 0		4 D E	1	9 5		2	4 C D		A	3	0 5

A12	A13	A14	A15	A16	A17	A18
1011001011000001	11100011010101011	01101	01011	110000	00	00000000000000
B 2 C 1	E 3 5 5	B	5 7	8	0	0 0 0 0

EX1	EX2	EX3	EX4	EX5	EX6	EX7	Vn	Rsv	CRC
0011001100110000	0	0000000000000000	0000000000000000	00000	00000	00000000	000001	000000	100010101010000001100010
C C C	0	0 0 0 0	0 0 0 0	0 0	0	0	0 1	0	2 2 A 8 1 8 8

A1-Message Type : Alert
A2-Country/Region Name : Japan
A3-Provider Identifier : FMMC
A4-Hazard Category and Type : Rainfall
A5-Severity : Severe
A6-Hazard Onset：Week Number　: Current week
A7-Hazard Onset：Time of the Week　: SUNDAY - 01:00 PM
A8-Hazard Duration　: 6H <= Duration < 12H
A9-Type of Library　: Country/region library
A10-Version of Library　: Library version#1
A11-Instruction library : "Keep away from Water area."
Target area : Area enclosed in the ellipse indicated by the following parameters

- A12-Ellipse Centre Latitude : 35.688
- A13-Ellipse Centre Longitude : 139.691
- A14-Ellipse Semi-Major Axis : 10.933(km)
- A15-Ellipse Semi-Minor Axis : 5.979(km)
- A16-Ellipse Azimuth : 45.00 (degrees)

A17-Type of Specific Settings　: B1-Improved Resolution of Main Ellipse
A18- Specific Settings

- Refined Latitude of Centre of Main Ellipse : 0.000000
- Refined Longitude of Centre of Main Ellipse : 0.000000000
- Refined Length of Semi-Major Axis : 1.000

Refined Length of Semi-Minor Axis : 1.000

EX1-Target Area Code : Shinjuku, Tokyo

EX2 to EX7 : Not used

(2) DCX message (information from local government)

Format	PAB	MT	SD	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11
BIN	01010011	10110	0000110000	01	001101111	00100	1010010	10	0	1001001100 1101	10	1	000	0000000000
HEX	5 3	B	0 6 0		4 D	E	4 A	5		2 4 C D		A		0 0 0

A12	A13	A14	A15	A16	A17	A18
1011001011000001	1110001101010101 1	01101	01011	110000	00	00000000000000
B 2 C 1	E 3 5 5	B	5 7	8	0	0 0 0 0

EX1	EX2	EX3	EX4	EX5	EX6	EX7	Vn	Rsv	CRC
00110011001100 00	0	1011001011000010	0000000000000000	01101	01011	1100000	000001	000000	111010110011101010010001
C C C	1	6 5 8 2	0 0 0 0	3	5	7 8	0 1	0 3	A C E A 4 4

A1-Message Type : Alert

A2-Country/Region Name : Japan

A3-Provider Identifier : Local Government

A4-Hazard Category and Type : Tropical Cyclone (Typhoon)

A5-Severity : Severe

A6-Hazard Onset : Week Number : Current week

A7-Hazard Onset : Time of the Week : SUNDAY - 01:00 PM

A8-Hazard Duration : 6H <= Duration < 12H

A9-Type of Library : Country/region library

A10-Version of Library : Library version#1

A11-Instruction library : Not specified (Reference EX2 to EX7.)

Target area : Area enclosed in the ellipse indicated by the following parameters

A12-Ellipse Centre Latitude : 35.688

A13-Ellipse Centre Longitude : 139.691

A14-Ellipse Semi-Major Axis : 10.933(km)

A15-Ellipse Semi-Minor Axis : 5.979(km)

A16-Ellipse Azimuth : 45.00 (degrees)

A17-Type of Specific Settings : B1-Improved Resolution of Main Ellipse

A18- Specific Settings

Refined Latitude of Centre of Main Ellipse : 0.000000

Refined Longitude of Centre of Main Ellipse : 0.000000000

Refined Length of Semi-Major Axis : 1.000

Refined Length of Semi-Minor Axis : 1.000

EX1-Target Area Code : Shinjuku, Tokyo

EX2-Evacuate Direction Type : "Leave the additional target area range."

Additional target area : Area enclosed in the ellipse indicated by the following parameters

EX3-Additional Ellipse Centre Latitude : 35.687
EX4-Additional Ellipse Centre Longitude : 139.689
EX5-Additional Ellipse Semi-Major Axis : 10.933(km)
EX6-Additional Ellipse Semi-Minor Axis : 5.979(km)
EX7-Additional Ellipse Azimuth : 45.00 (degrees)

5.3.2. DCX message (J-Alert)

Format	PAB	MT	SD	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11
BIN	10011010	101100	0001100000	01	001101111	00010	0001000	10	0	00001011010001	00	1	000	1000000011
HEX	9 A B 0 6 0 4 D E 2 1 1 0 2 D 1 4 2 0 3													

A12	A13	A14	A15	A16	A17	A18
0000000000000000	00000000000000000	00000	00000	000000	00	000000000000000
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0						

EX8	EX9	EX10	Vn	Rsv	CRC
0	0000000000000000000000000000000000000000000100110000000000000000000	000	000001	000000	010011001111000001111100
0 0 0 0 0 0 0 0 0 0 0 4 C 0 0 0 0 0 1 0 1 3 3 C 1 F 0					

A1-Message Type : Alert

A2-Country/Region Name : Japan

A3-Provider Identifier : FDMA

A4-Hazard Category and Type : Missile Attack

A5-Severity : Severe

A6-Hazard Onset：Week Number　: Current week

A7-Hazard Onset：Time of the Week　: MONDAY - 00:00 PM

A8-Hazard Duration　: Duration=00

A9-Type of Library　: Country/region library

A10-Version of Library　: Library version#1

A11-Instruction library : "Move to. Underground."

A12 to A18 : Not used

EX8-Target Area Code List Type : Prefecture code

EX9-Target Area Code List : Hokkaido, Aomori, Akita

5.4. Transmission Overview

This section shows examples of message transmission for DCX message (L-Alert), DCX message (J-Alert), and DCX message (information from local government). Message transmission for DCX message (information from organizations outside Japan) is undecided.

Basically, a received message may be handled according to the receiver's unique criteria.

5.4.1. DCX message (L-Alert)/ DCX message (information from local government)

Examples of MT44 message transmission for L-Alert and information from local government are shown below.

(1) Clearing alerts using a DCX message (A1-Message Type="All Clear")

Alerts are cleared when a DCX message in which A1-Message Type is set to "All Clear" is received.

Figure 5.4-1 shows an example of clearing alerts using All Clear.

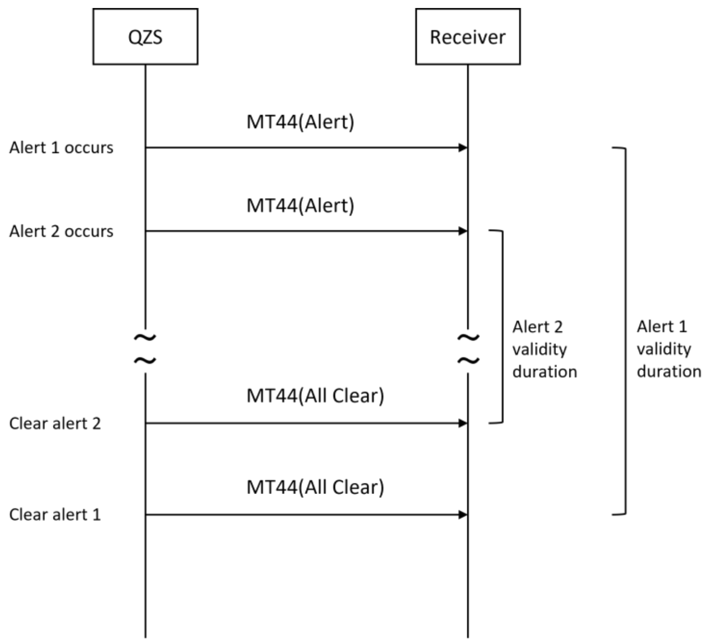


Figure 5.4-1 Clearing Alerts Using All Clear

(2) Clearing alerts using the A8-Hazard Duration

When a DCX message in which A1-Message Type is set to "All Clear" has yet to be received, alerts are cleared according to the validity duration indicated by the A8-Hazard Duration. See 5.11for details.

Figure 5.4-2 shows an example of clearing alerts according to the Hazard Duration.

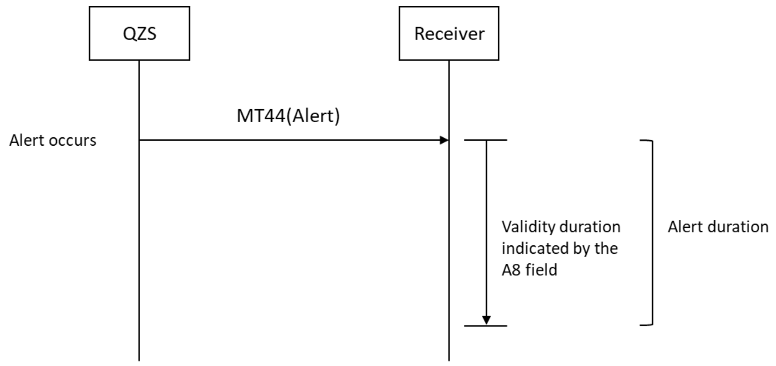


Figure 5.4-2 Clearing Alerts According to the Hazard Duration

When a DCX message in which A1-Message Type is set to "All Clear" has yet to be received and when the A8 field is set to "00" (Unknown), alerts received a week ago or earlier are cleared.

5.4.2. DCX message (J-Alert)

DCX message (J-Alert) generally supports only the transmission of DCX messages (A1-Message Type="Alert"). It does not support clearing alerts using DCX messages (A1-Message Type="All Clear").

An example of message transmission is shown below.

(Example) Example of message transmission in case of ballistic missile launch

Figure 5.4-3 shows an example of message transmission.

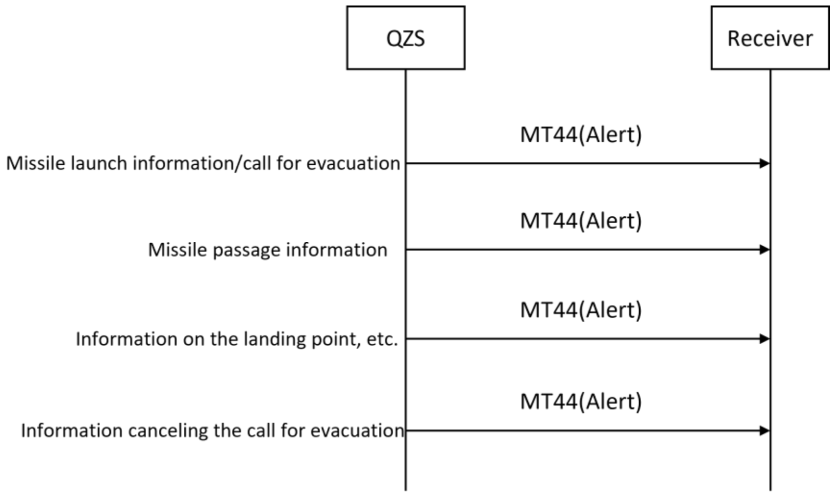


Figure 5.4-3 Example of Message Transmission for DCX message (J-Alert)

As shown in the example above, messages to cancel a call for evacuation are also transmitted as a message in which the A1-Message Type is set to "Alert."

It is therefore not possible to determine whether the message is a call for evacuation or a cancellation of the call from the A1-Message Type. To identify whether the message is a call to evacuate or a cancellation of the evacuation call, refer to the contents provided in the A11-Instruction library.

If a DCX message (A1-Message Type="Alert") is a false alert, a DCX message (A1-Message Type="All Clear") is transmitted.

Figure 5.4-4 shows an example of message transmission in case of a false alert.

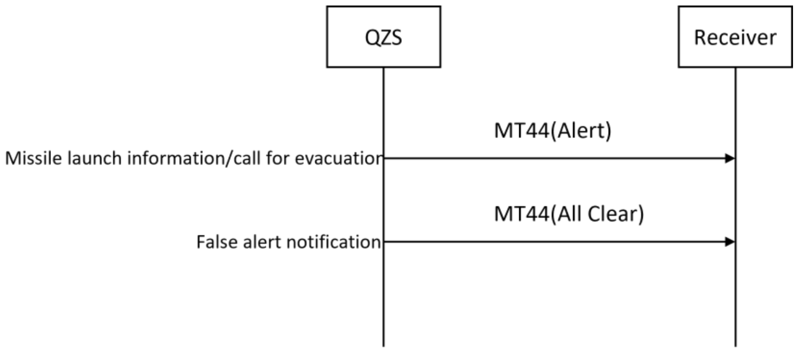


Figure 5.4-4 Example of Message Transmission for DCX message (J-Alert) (False Alert)

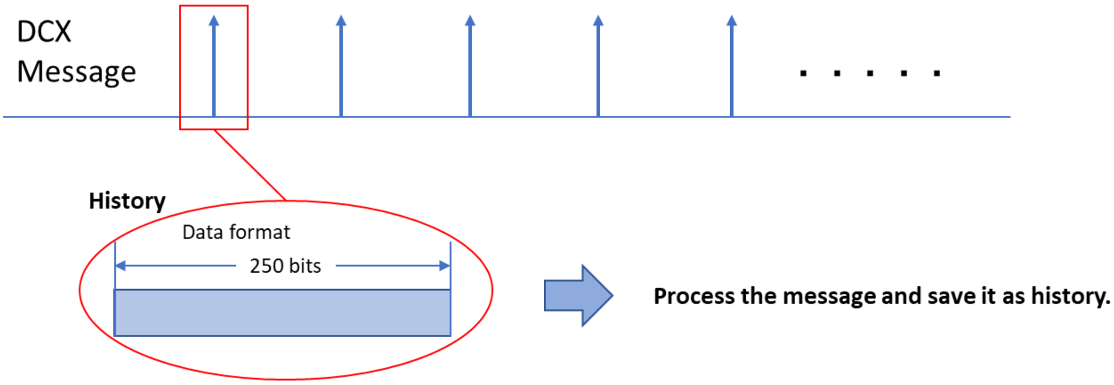
5.5. Message Duplication Checking Method

Since a message is transmitted repeatedly (typically twice; this can be changed by setting the QZSS parameter), a duplication check is necessary.

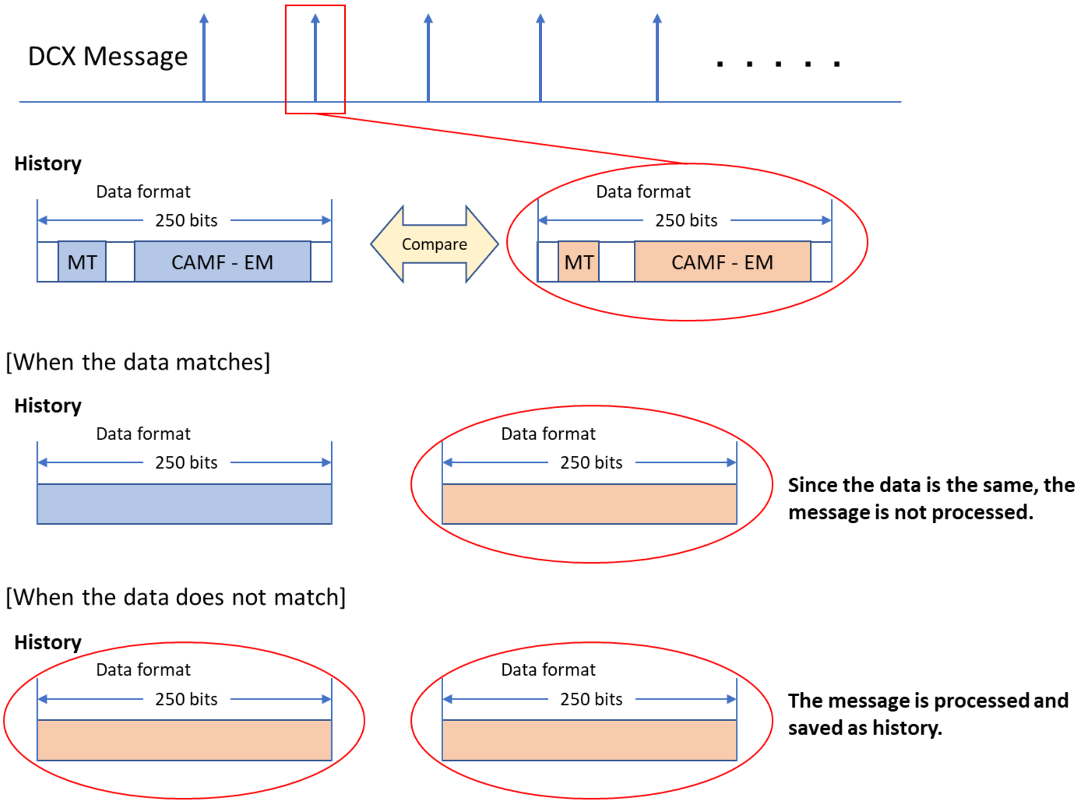
When performing duplication check, it is necessary to note that the same message may not be received consecutively. As described in the transmission order outlined in section 4.1.4, higher priority messages may interrupt lower priority messages. Therefore, the same message may not always be received twice in a row.

An example of checking method is described below.

Step (1) Save the first received message as history.



Step (2) Compare the saved history and the next message to check whether MT (Message Type ID) and CAMF to EM (Extended Message) match.



5.6. SD - Satellite Designation Field Usage Method

5.6.1. Overview

The SD field notifies the receiver about the service type indicating which QZSS satellite transmits MT44 for Japan and which QZSS satellite transmits MT44 for use outside Japan, as well as about the MT44 transmission status indicating which QZSS satellite is transmitting MT44. The receiver selects the QZSS satellite to capture, based on the service type and MT44 transmission status.

- Service type (SDMT="0")
  When the SDMT field is set to "0", the SDM field shows the service type. This is a bit flag indicating the service type of each QZSS satellite.
  For example, when the nine bits of the SDM field are "010000100", only the two satellites corresponding to Bit_2 and Bit_7 are of the service type for use outside Japan. For the seven satellites corresponding to the remaining bits, Bit_1, Bit_3 to Bit_6, Bit_8, and Bit_9, the service type for Japan is specified.
- MT44 transmission status (SDMT="1")
  When the SDMT field is set to "1", the SDM field shows the MT44 transmission status. This is a bit flag indicating the MT44 transmission status of each QZSS satellite.
  For example, when the nine bits of the SDM field are "011100100", it indicates that only the four satellites corresponding to Bit_2 to Bit_4, and Bit_7 are transmitting the MT44 message. Note that the MT44 transmission status does not change unless a fault occurs or the configuration of QZSS satellites transmitting MT44 is changed. Therefore, it is transmitted less frequently than the service type (once a minute or so).

A capture target satellite list is created by combining the data obtained from the service type and MT44 transmission status. This list shows the candidate satellites to be captured by the receiver. The same MT44 is transmitted from satellites with the same service type. Therefore, even if the capture target satellite list has multiple satellites but not all the satellites can be captured for reasons such as the limit to the number of receive channels, the receiver may select the satellite to capture using its unique metrics.

In the case of the receiver for Japan, the capture target satellite list includes the satellites for which the service type is set to 0 and the MT44 transmission status is set to 1 (Table 5.6-1).

Table 5.6-1 Example of a Capture Target Satellite List Created for Japan

SDM Field	Bit_1	Bit_2	Bit_3	Bit_4	Bit_5	Bit_6	Bit_7	Bit_8	Bit_9
Service Type	0	1	0	0	0	0	1	0	0
MT44 Transmission Status	0	1	1	1	0	0	1	0	0
Capture Target Satellite List			●	●					

In the case of the receiver for use outside Japan, the capture target satellite list includes the satellites for which the service type is set to 1 and the MT44 transmission status is set to 1 (Table 5.6-2).

Table 5.6-2 Example of a Capture Target Satellite List Created for Use Outside Japan

SDM Field	Bit_1	Bit_2	Bit_3	Bit_4	Bit_5	Bit_6	Bit_7	Bit_8	Bit_9
Service Type	0	1	0	0	0	0	1	0	0
MT44 Transmission Status	0	1	1	1	0	0	1	0	0
Capture Target Satellite List		●					●		

Table 5.6-3 shows the PRNs and QZSS satellites to assign to the SDM field.

Table 5.6-3 PRNs and QZSS Satellites to Assign to the SDM Field

SDM Field	PRN	QZSS Satellites	Remark
Bit_1	183	QZS1	
Bit_2	184	QZS2	
Bit_3	185	QZS4	
Bit_4	186	QZS1R	
Bit_5	-	-	Not assigned
Bit_6	-	-	Not assigned
Bit_7	189	QZS3	
Bit_8	-	-	Not assigned
Bit_9	-	-	Not assigned

5.6.2. Example of the Transmission Sequence

Figure 5.6-1 shows an example of the transmission sequence for creating the capture target satellite list.

This example assumes that there are two receivers, one for Japan and one for use outside Japan, along with only one L1S receive channel. Four satellites, QZS1R, QZS2, QZS3, and QZS4, are transmitting MT44. By default, the service type for Japan is specified for QZS2 and QZS3 and the service type for use outside Japan is specified for QZS1R and QZS4. It is assumed that, after the initial start, the receiver has captured QZS1R as a tentative capture target. (The receiver may capture another QZSS satellite according to its unique criteria.)

The capture target satellite is determined when the two values of the service type and MT44 transmission status are received. Since the receiver (for Japan) includes two satellites, QZS2 and QZS3, in the capture target satellite list, it changes the capture target from QZS1R to QZS2. (The receiver may change to QZS3 according to its unique criteria.) While the receiver (for use outside Japan) includes two satellites, QZS1R and QZS4, in the capture target satellite list, there is no need to change the capture target because it has already captured QZS1R. (The receiver may change to QZS4 according to its unique criteria.)

If a change is made to the service type, the QZSS satellites transmit the message with the value of the service type changed. Since the receiver (for Japan) includes two satellites, QZS1R and QZS3, in the capture target satellite list, it changes the capture target from QZS2 to QZS1R. (The receiver may change to QZS3 according to its unique criteria.) Since the receiver (for use outside Japan) includes two satellites, QZS2 and QZS4, in the capture target satellite list, it changes the capture target from QZS1R to QZS2.

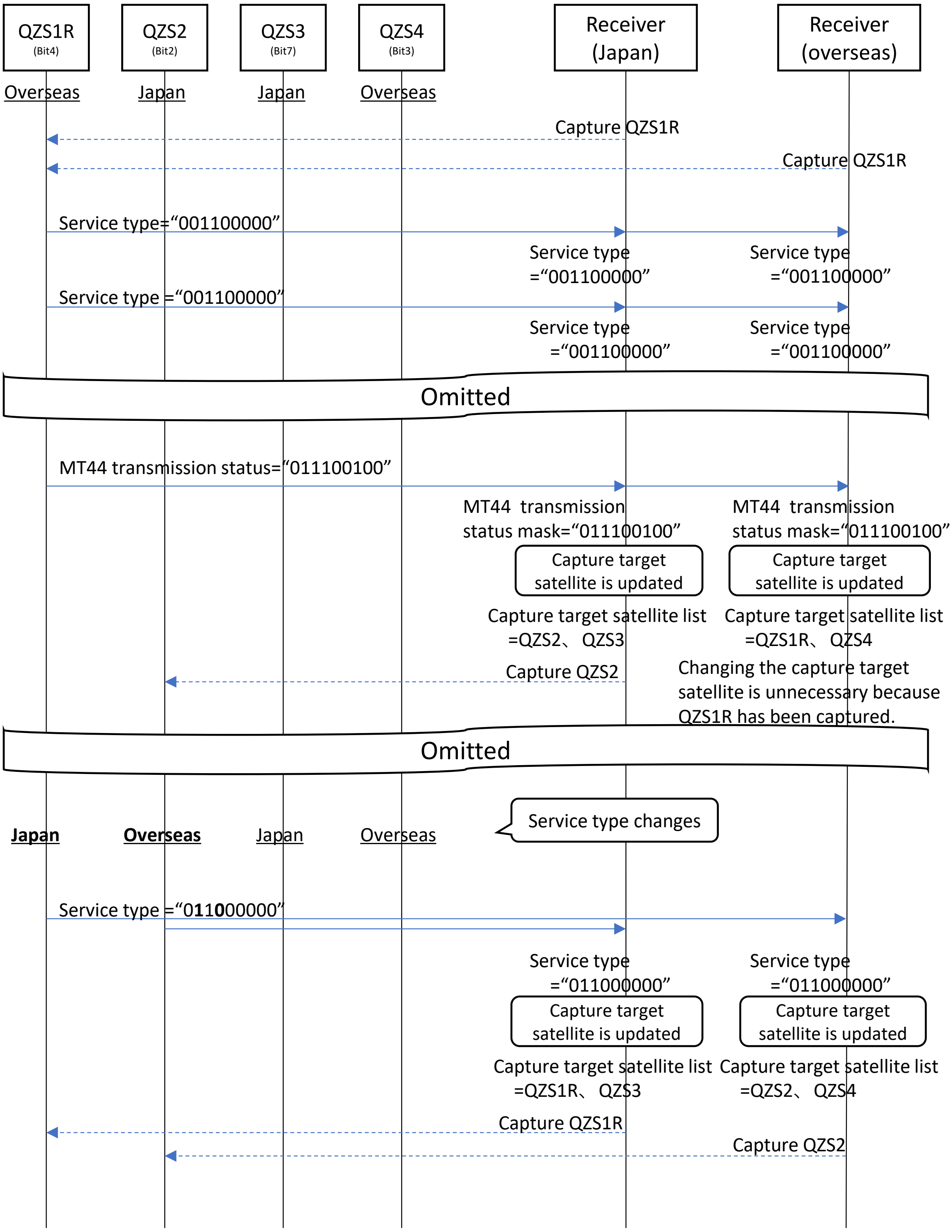


Figure 5.6-1 Example of the SD Field Transmission Sequence

5.6.3. Example of the User Algorithm

Figure 5.6-2 shows an example of the user algorithm used for the receiver to process the SD field. This is a mere example, and the receiver may process the field using its unique flowchart.

Immediately after the start, the receiver captures a QZSS satellite to receive MT44. While this example assumes that the receiver captures QZS1R, the receiver may capture any other QZSS satellite according to its unique criteria. After that, it receives MT44 at regular intervals.

Upon receiving MT44, the receiver performs the specified application processing, processes the SD field data, and creates a capture target satellite list.

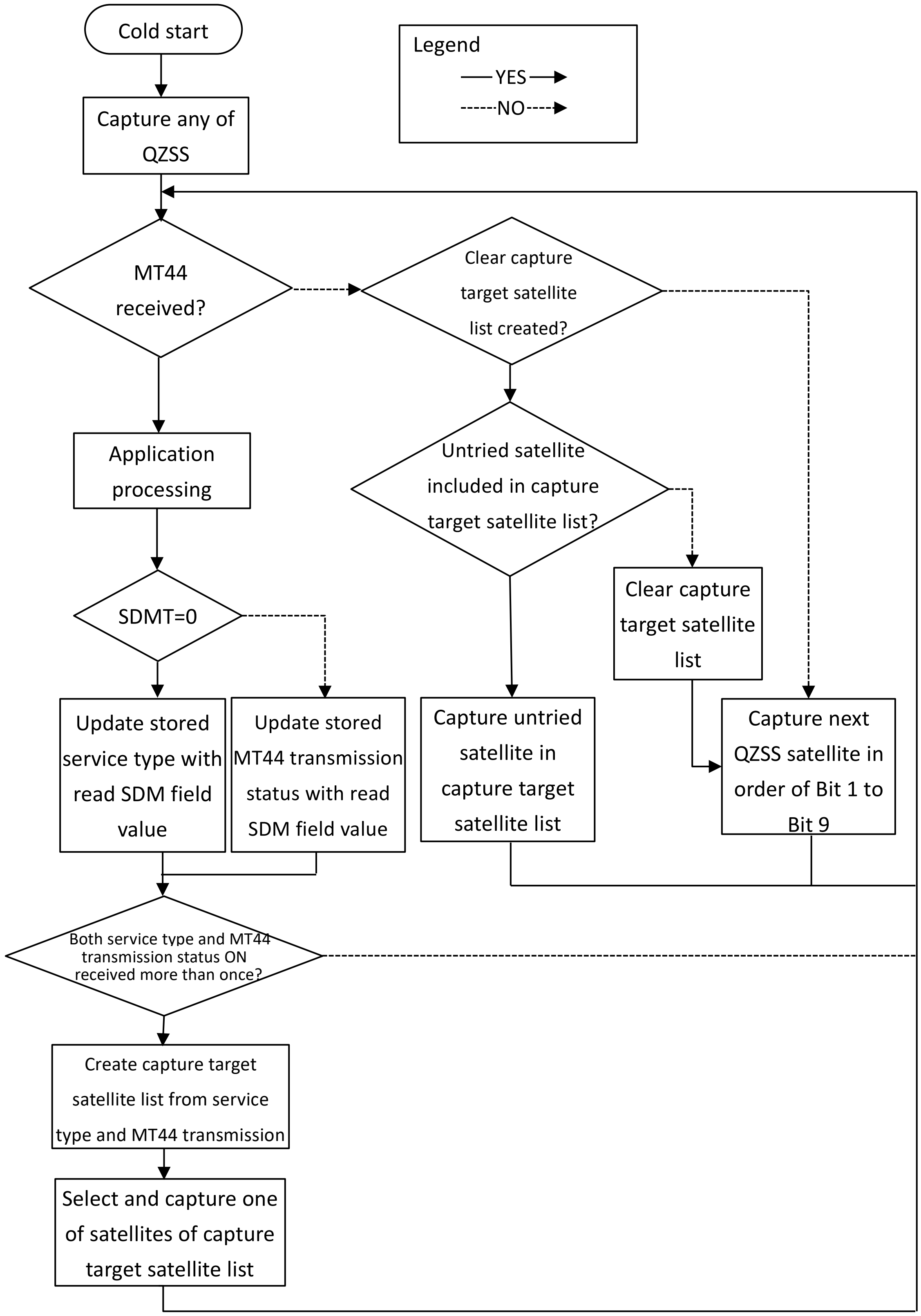


Figure 5.6-2 Example of the Algorithm for Reading the SD Field (Flowchart)

5.7. Method for Determining Types of Information Transmitted as MT44

DCX message type is determined by referencing the A2-Country/Region Name and A3-Provider Identifier.

・DCX message (information from organizations outside Japan)
A2 = Other than "001101111" (Japan)

・DCX message (L-Alert)
A2 = "001101111" (Japan) and A3 = "00001" (FMMC)

・DCX message (J-Alert)
A2 = "001101111" (Japan) and A3 = "00010" (FDMA) or "00011" (Related Ministries)

・DCX message (information from local government)
A2 = "001101111" (Japan) and A3 = "00100" (Local Government)

・Other than the above
When A2 is "001101111" (Japan) but A3 is other than "00001", "00010", "00011", or "00100", the message is not the intended information and is therefore discarded.

Figure 5.7-1 shows DCX message type determination flowchart.

The SD field needs to be processed regardless of whether the message is processed or discarded in the application processing. See 5.6.3

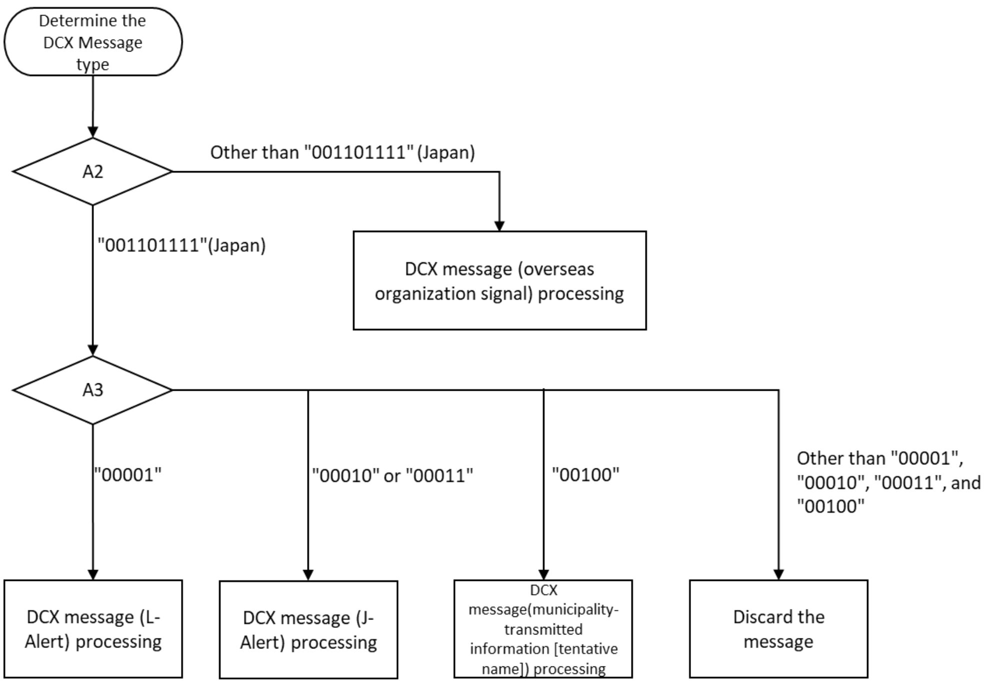


Figure 5.7-1 DCX Message Type Determination Flowchart

5.8. Target Area Determination Method

The target area is represented either by the area enclosed in the ellipse indicated by the A12 to A16 fields or by the Target Area Code in an Extended Message.

5.8.1. Determining the target area of a DCX message (L-Alert)

The target area is represented in the following two fields.

・Area enclosed in the ellipse indicated by the A12 to A16 fields

・Citie, town and village indicated by the EX1 field

When the A12 to A16 fields are all "0", the EX1 field is referenced.

Figure 5.8-1 shows the target area determination flowchart.

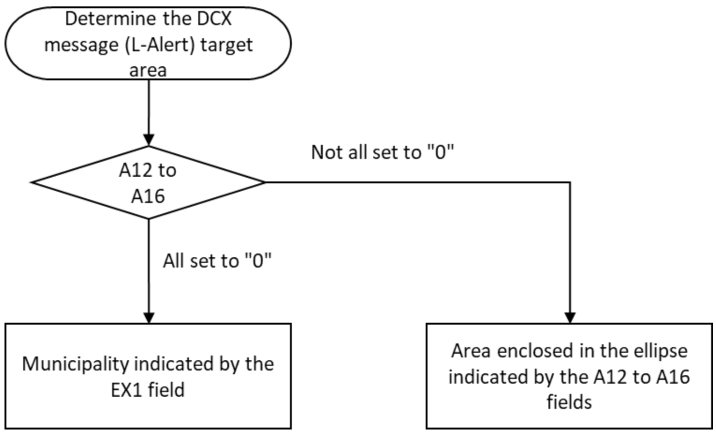


Figure 5.8-1 Target Area Determination Flowchart for DCX message (L-Alert)

5.8.2. Determining the target area of a DCX message (J-Alert)

The target area is represented in the Extended Message fields: EX8（Target Area Code List Type） and EX9（Target Area Code List）. The A12 to A16 fields are not used (A12 to A16 fields are all "0").

Depending on the value of EX8, the contents of EX9 differ.

・When EX8 = "0" (prefecture)

EX9 is a prefecture code shown in Table 4.2-25, and the target area is the prefecture indicated by the prefecture code.

・When EX8 = "1" (cities, towns and villages)

EX9 is a cities, towns and villages code shown in Table 4.2-21, and the target area is the cities, towns and villages indicated by the cities, towns, and villages code.

Figure 5.8-2 shows the target area determination flowchart.

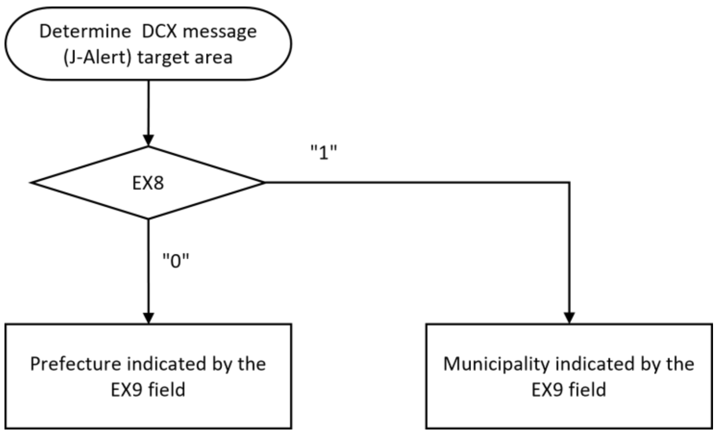


Figure 5.8-2 Target Area Determination Flowchart for DCX message (J-Alert)

5.9. Ellipse Inside/Outside Determination

The ellipse inside/outside determination method is described below that is used to determine whether or not given coordinates are within the range with respect to the target area indicated by the ellipse parameters of A12 to A16 (Target area - Ellipse) shown in Table 4.2-3 or EX3 to EX7 (Additional Ellipse) shown in Table 4.2-20.

Since the above-mentioned two types of ellipse parameters differ only in the value range and accuracy, the inside/outside determination can be made using the same method. This section describes the procedure using only A12 to A16 (Target area - Ellipse).

■ Premise

The parameters are as follows. See Figure 5.9-1.

■ Ellipse − Centre latitude (A12): lat_0 (unit: degrees) − Centre longitude (A13): lon_0 (unit: degrees) − Semi-Major Axis (A14): a (unit: km) − Semi-Minor Axis (A15):b (unit: km) − Azimuth (A16):θ (unit: degrees) − Center coordinates: E_0 (lat_0, lon_0)	■ Target points for inside/outside determination − Latitude: lat_{p0} (unit: degrees) − Longitude: lon_{p0} (unit: degrees) − Coordinates: P_0 (lat_{p0}, lon_{p0})

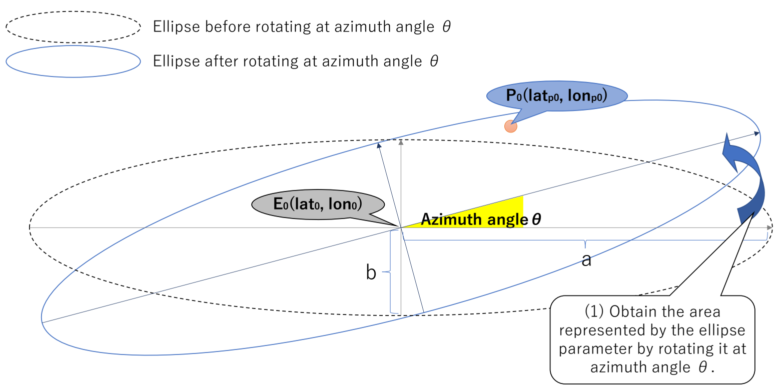


Figure 5.9-1 Ellipse Inside/Outside Determination – Premise

(1) Convert the ellipsoidal coordinate system to a local Cartesian coordinate system.

Convert the target point coordinates P_0 (lat_{p0}, lon_{p0}) in the ellipsoidal coordinate system and the ellipsoidal center coordinates E_0 (lat_0, lon_0) to the ENU (East-North-Up) local Cartesian coordinate system, using E_0 as the origin. This results in $P_1(x_{p1}, y_{p1})$, $E_1(x_1, y_1) = (0, 0)$, respectively. The WGS84 geodetic system is employed.

An example of the calculation procedure is shown below. It should be noted that the calculations are based on an ellipsoidal height of 0.

1. Convert ellipsoidal coordinates P_0 and E_0 to the ECEF coordinate system using Equation 5.9.1. Ellipsoidal coordinates Q (lat_Q, lon_Q) are converted to the ECEF coordinate system. 'a_e' represents the semi-major axis of the Earth's ellipsoid (WGS84), 'f' denotes the flattening of the Earth's ellipsoid, and 'e' stands for the eccentricity.

Equation 5.9.1 Converting Geodetic Coordinates to ECEF Coordinates

$$ECEF(Q) = \begin{bmatrix} Q_x \\ Q_y \\ Q_z \end{bmatrix} = \begin{bmatrix} N \cos\varphi_Q \cos\lambda_Q \\ N \cos\varphi_Q \sin\lambda_Q \\ N(1-e^2)\sin\varphi_Q \end{bmatrix}$$

$$\varphi_Q = \frac{lat_Q \cdot \pi}{180}$$

$$\lambda_Q = \frac{lon_Q \cdot \pi}{180}$$

$$N = \frac{a_e}{\sqrt{1 - e^2\left(\sin\varphi_Q\right)^2}}$$

$$e^2 = 2f - f^2$$

$$a_e = 6378137$$

$$f = \frac{1}{298.257223563}$$

2. Using Equation 5.9.2Equation 5.9.2, compute the position vector $\overrightarrow{E_0P_0}$, representing the relative position from E_0 to P_0.

Equation 5.9.2 Computation of the Position Vector

$$\overrightarrow{E_0P_0} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} P_{0_x} - E_{0_x} \\ P_{0_y} - E_{0_y} \\ P_{0_z} - E_{0_z} \end{bmatrix}$$

3. Taking E_0 as the origin, convert the vector $\overrightarrow{E_0P_0}$ to the ENU local coordinate system using Equation 5.9.3

Equation 5.9.3 Conversion from Position Vector to ENU Local Coordinate System

$$ENU(P_0) = \begin{bmatrix} P_{0_e} \\ P_{0_n} \\ P_{0_u} \end{bmatrix} = \begin{bmatrix} -x\sin\lambda_{E_0} + y\cos\lambda_{E_0} \\ -x\sin\varphi_{E_0}\cos\lambda_{E_0} - y\sin\varphi_{E_0}\sin\lambda_{E_0} + z\cos\varphi_{E_0} \\ x\cos\varphi_{E_0}\cos\lambda_{E_0} - y\cos\varphi_{E_0}\sin\lambda_{E_0} + z\sin\varphi_{E_0} \end{bmatrix}$$

4. Interpret and handle the ENU coordinates (e, n) as plane coordinates (x, y) according to Equation 5.9.4.

Equation 5.9.4 Interpreting ENU Local Coordinates as Plane Coordinates

$$x_{p1} = P_{0_e}$$

$$y_{p1} = P_{0_n}$$

(2) Rotate P_1 backward at azimuth angle θ so that it is parallel to the ellipse before rotating at azimuth angle θ.

Rotate coordinates P_1 backward at azimuth angle θ, with the center being coordinates E_1, which are the center point of the ellipse, in order to obtain coordinates P_2 (x_2, y_2). See Equation 5.9.5 and Figure 5.9-2.

Equation 5.9.5 Rotating P_1 Backward by θ with E_1 Being the Center

$$x_{p2} = (x_{p1} - x_1) \times \cos(-\theta) - (y_{p1} - y_1) \times \sin(-\theta) + x_1$$
$$y_{p2} = (x_{p1} - x_1) \times \sin(-\theta) + (y_{p1} - y_1) \times \cos(-\theta) + y_1$$

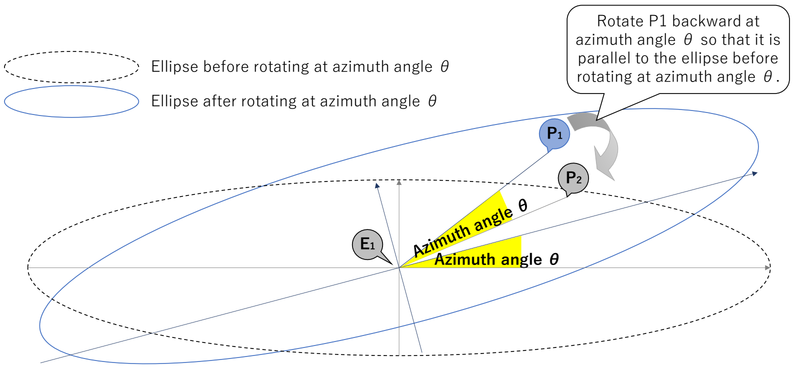


Figure 5.9-2 Ellipse Inside/Outside Determination - Rotating P_1 Backward by θ with E_1 Being the Center

(3) Calculate the ellipse equation (standard) for inside/outside determination.

Apply major axis radius a, minor axis radius b, and coordinates P_2 (x_{p2}, y_{p2}) to the ellipse equation (standard). The coordinates are determined to be inside the ellipse when right side value is less than 1, or outside the ellipse when this value is more than 1. See Equation 5.9.6 and Figure 5.9-3.

Equation 5.9.6 Application to the Ellipse Equation (Standard)

$$Ellipse\ equation\ (standard) = \frac{{x_{p2}}^2}{a^2} + \frac{{y_{p2}}^2}{b^2}$$

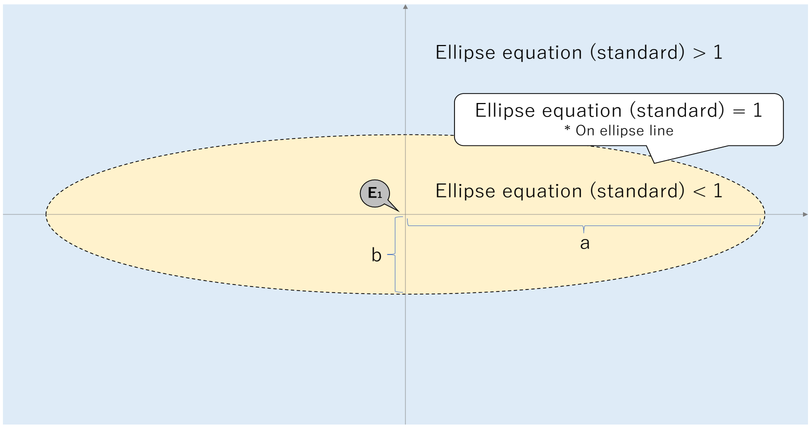


Figure 5.9-3 Ellipse Inside/Outside Determination - Range Indicated by the Ellipse Equation Result

5.10. Referencing the Evacuation Action (DCX message (L-Alert), DCX message (J-Alert), DCX message (information from local government)

DCX message has two types of evacuation action instruction as shown below:

1 : A11-Instruction library

This field indicates the evacuation actions listed in Table 4.2-15. This instruction is available when the message is DCX message (L-Alert) , DCX message (J-Alert), and DCX message (information from local government). When all the bits of A11 are set to "0", this field indicates no instruction.

2 : EX2-Evacuate Direction Type, and EX3 to EX7 Additional Ellipse

These fields specify in more detail the area from which people are instructed to "Leave" or to which they are instructed to "Head" using the additional ellipse. This instruction is available when the message is DCX message (information from local government). When all the bits of EX2 and EX3 to EX7 are set to "0", EX2 and EX3 to EX7 indicate no instruction.

Both types of instruction can indicate the area to evacuate from or head to, but the EX2 and EX3 to EX7 fields specify this area more precisely than the A11 field. "Instructions by the EX2 and EX3 to EX7 fields are available only when the message is a DCX message (information from local government). In such cases, there are instances where both types are broadcast, and instances where only one type is broadcast." When both types are broadcast, it is recommended to follow the combined instructions. When the message is a DCX message (L-Alert) or a DCX message (J-Alert), only instruction by the A11 field is available.

5.11. Using the A8-Hazard Duration

The A8-Hazard Duration indicates the expected validity duration of a DCX message (in hours (H)).

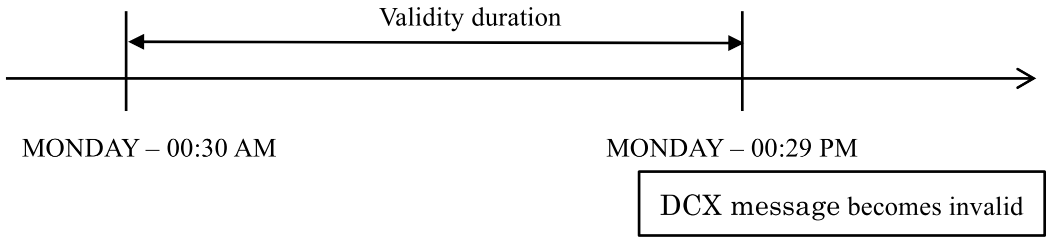
This field consists of two bits to indicate the following four patterns.

00: Duration Unknown
01: Duration < 6H
10: 6H ≤ Duration < 12H
11: 12H ≤ Duration < 24H

The time the duration starts is specified by the A7-Hazard Onset : Time of the Week. (A7 specifies UTC time.)

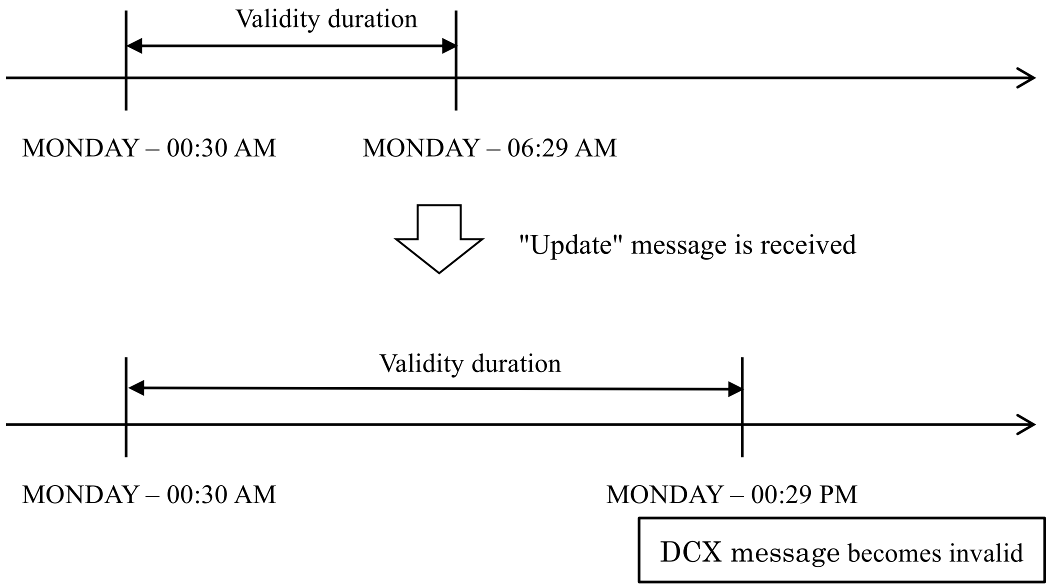
An example of using the A8 field is shown below.

(Example) When A7 = " 00000000011111" (MONDAY - 00:30 AM) and A8 = "10"



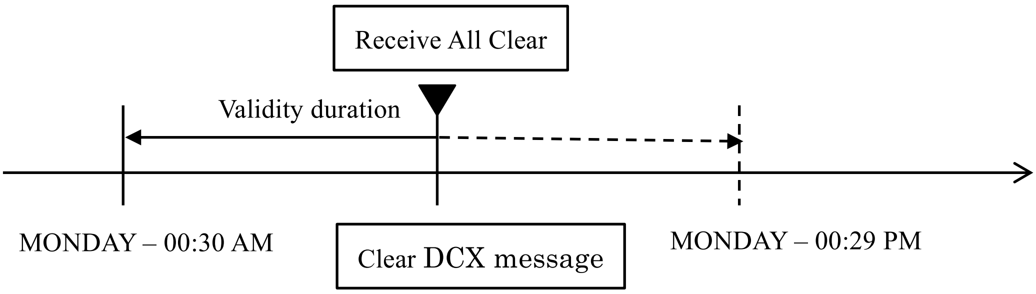
In some cases, the "Update" message (A1-Message Type = "10") may update the validity duration.

(Example) When the "Update" message in which A8 is set to "10" is received after reception of the "Alert" message in which A7 is set to "00000000011111" (MONDAY - 00:30 AM) and A8 is set to "01"



If the "All Clear" message (A1-Message Type = "11") is received during the validity duration, the corresponding DCX message is cleared.

(Example) When the "All Clear" message is received during the validity duration after reception of the "Alert" message in which A7 is set to " 00000000011111" (MONDAY - 00:30 AM) and A8 is set to "10"



5.12. Geo-Fencing

To notify an alert only to people threatened or affected by a situation, the EWSS concept applies the geo-fencing principle. By comparing its own geographical position with both the country / region code and the geodetic coordinates of the ellipse encoded in the received EWM, the receiver is able to determine whether to notify the alert to the user or not. In other words, the receiver notifies the alert if the following condition is true:

$$Receiver\ position\ \in (\ Country \cap Ellipse\ )$$

This double condition allows the receiver to resolve the issue of cross-border alerts, when the ellipse coded for the alert area crosses the border with a neighbouring country. To avoid notifying people located across the border with an alert that does not originate from their own country, the receiver reads the country code in the CAMF: if the user is located within the ellipse but not in the country that has emitted the alert (i.e. the user is located in the parts of the ellipse that has crossed the border), the receiver shall not notify the user.

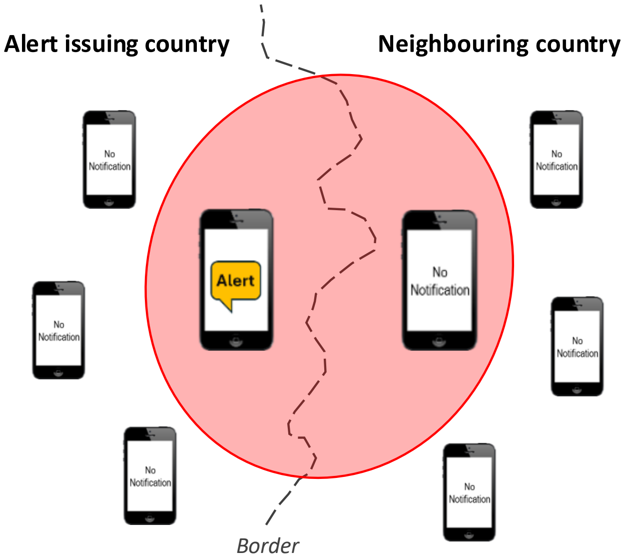


Figure 5.12-1 Geo-fencing principle